

CollegeConnect

A Project Report for Industrial Training and Internship

submitted by

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In the partial fulfilment of the award of the degree of

BCA

in the

Department Of Computer Application

Of

Chandannagar Institute Of Management & Technology



At

Ardent Computech Pvt. Ltd.





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Signature Of the Supervisor

Date:

Name Of the Project Supervisor:





BONAFIDE CERTIFICATE

Certified that this project work was carried out under my supervision

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EXAMINERS



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1. COMPANY PROFILE

ARDENT (Ardent Computech Pvt. Ltd.), formerly known as Ardent Computech Private Limited, is an ISO 9001:2015 certified Software Development and Training Company based in India. Operating independently since 2003, the organization has recently undergone a strategic merger with ARDENT Technologies, enhancing its global outreach and service offerings.

ARDENT Technologies

ARDENT Technologies delivers high-end IT services across the UK, USA, Canada, and India. Its core competencies lie in the development of customized application software, encompassing end-to-end solutions including system analysis, design, development, implementation, and training. The company also provides expert consultancy and electronic security solutions. Its clientele spans educational institutions, entertainment companies, resorts, theme parks, the service industry, telecom operators, media, and diverse business sectors.

ARDENT Collaborations

ARDENT Collaborations, the Research, Training, and Development division of ARDENT (Ardent Computech Pvt. Ltd.), offers professional IT-enabled services and industrial training programs. These are tailored for freshers and professionals from B.Tech, M.Tech, MBA, MCA, BCA, and MSc backgrounds. ARDENT (Ardent Computech Pvt. Ltd.) provides Summer Training, Winter Training, and Industrial Training to eligible candidates. High-performing students may qualify for stipends, scholarships, and additional benefits based on performance and mentor recommendations.

Associations and Accreditations

ARDENT (Ardent Computech Pvt. Ltd.) is affiliated with the National Council of Vocational Training (NCVT) under the Directorate General of Employment & Training (DGET), Ministry of Labour & Employment, Government of India. The institution upholds strict quality standards under ISO 9001:2015 certification and is dedicated to bridging the gap between academic knowledge and industry skills through innovative training programs.

2. INTRODUCTION

In today's interconnected world, effective communication and collaboration are paramount, especially within the academic sphere. While technology has transformed many aspects of education, the direct, seamless interaction between different educational institutions often remains fragmented. This project, CollegeConnect, aims to bridge this gap by creating a comprehensive and interactive platform designed to foster efficient and streamlined communication and collaboration among colleges.

Powered by the MERN stack, CollegeConnect is engineered to provide a centralized and user-friendly hub for various inter-college interactions. It enables institutions to securely register, manage their profiles, and engage in a wide range of activities including sharing valuable resources, coordinating events, participating in public discussions, and discovering other institutions within the network. By leveraging modern web technologies and a well-structured architecture, the platform ensures an engaging and effective environment for inter-institutional exchange.

CollegeConnect not only streamlines administrative and academic communication between educators and institutions but also promotes knowledge sharing, collaborative initiatives, and networking among the academic community. It is designed to be an indispensable tool for colleges looking to enhance their reach, foster partnerships, and create a more connected and efficient educational ecosystem.

2A.OBJECTIVE

The primary objective of CollegeConnect is to revolutionize inter-institutional engagement by providing a flexible, accessible, and comprehensive digital platform that streamlines communication and fosters collaboration among diverse colleges and universities. The platform aims to bridge critical gaps in direct institutional interaction by offering a wide range of secure communication channels, centralized resource management, and tools for collaborative initiatives.

Ultimately, CollegeConnect seeks to empower educational institutions by facilitating seamless knowledge exchange, promoting joint academic and administrative ventures, and enhancing networking opportunities within the broader academic community—enabling them to succeed in their collective endeavors and contribute to a more connected educational ecosystem.

2B.SCOPE

Our project, CollegeConnect, focuses on creating a comprehensive web-based application designed to facilitate seamless and secure communication, collaboration, and resource sharing among various educational institutions.

1. College Registration & Login:

Provides secure sign-up/login for college representatives, including password management (forgot/reset) and Google Sign-in authentication.

2. College Profile Management:

Enables registered colleges to create, view, and update detailed institutional profiles, including logo uploads and comprehensive contact/location information.

3. Resource Sharing Module:

Allows colleges to upload, manage, and share categorized documents (Official, Event, Reports & Academic, Administrative), with features for viewing details and secure downloading.

4. Event Management Module:

Facilitates the creation, management, and viewing of inter-college events like workshops, seminars, and fests.

5. Forum & Discussion Board:

Offers a dedicated platform for colleges to initiate public discussions through posts and includes functionality for mentioning other colleges.

6. Colleges Directory:

Provides a browsable and searchable directory of all registered colleges, enabling users to explore and view their detailed public profiles.

7. Real-time Notification System:

Integrates an in-app notification system to alert colleges about new resources, events, forum posts, and mentions for timely awareness.

8. Responsive User Interface & Theme Customization:

Developed with React.js, Tailwind CSS, and Shadcn UI for an intuitive, visually appealing, and fully responsive experience across devices. Includes light and dark mode theming options for user preference, applied consistently throughout the application.

3. SYSTEM ANALYSIS

3A.IDENTIFICATION OF NEED

System analysis is a crucial phase in the development of our CollegeConnect project, involving creating an efficient and user-friendly platform for inter-institutional communication and collaboration. With rapid digital transformation, traditional inter-institutional communication methods remain fragmented and inefficient, hindering coordination and collaboration..

Key Needs Identified:

1.Streamlined Inter-institutional Communication:

- Colleges require direct, efficient, and centralized channels for academic, administrative, and collaborative communication, overcoming current reliance on disparate tools.

2.Centralized Resource Sharing:

- There is a need for a secure and organized platform for colleges to share official documents, event materials, academic content, and administrative guidelines, fostering knowledge transfer.

3.Enhanced Event Visibility and Coordination:

- Colleges need a platform to effectively announce their events to a wider academic network and discover events hosted by other institutions.

4.Fostering Inter-College Collaboration & Discussion:

- A dedicated space is required for colleges to initiate public discussions and engage in collaborative dialogues, including the ability to mention specific institutions.

5.Efficient College Discovery:

- Institutions need a comprehensive, searchable directory to easily identify, browse, and connect with other colleges for strategic partnerships.

3B.FEASIBILITY STUDY

The feasibility study affirms CollegeConnect's viability as a valuable platform for inter-institutional communication and collaboration.

Technically, the project is highly feasible, leveraging MERN stack technologies (React.js, Node.js/Express.js, MongoDB/Mongoose), alongside Tailwind CSS, Shaden UI, and critical modules like Passport.js, Nodemailer, and Multer. These are widely adopted and scalable solutions. Operationally, CollegeConnect is designed for user-friendliness and efficiency, directly addressing the identified needs of colleges for streamlined communication, resource sharing, and event coordination. Economically, its potential lies in fostering institutional partnerships and enhancing academic operations, offering significant value creation. Legally, the project is viable with due consideration for data protection and content licensing. The overall development timeline is achievable within standard academic project parameters. CollegeConnect is thus a realistic, impactful, and sustainable solution within the academic technology landscape.

3C.WORKFLOW

This Document plays a vital role in the development life cycle (SDLC) as it describes the complete requirements of the system. It is meant for use by the developers and will be the basic during the testing phase. Any changes made to the requirements in the future will have to go through a formal change approval process.

The Waterfall Model was the first Process Model to be introduced. It is also referred to as a linear-sequential life cycle model. It is very simple to understand and use. In a waterfall model, each phase must be completed before the next phase can begin and there is no overlapping in the phases.

The waterfall model is the earliest SDLC approach that was used for software development. The waterfall Model illustrates the software development process in a linear sequential flow; hence it is also referred to as a linear-sequential life cycle model. This means that any phase in the development process begins only if the previous phase is complete. In the waterfall model phases do not overlap.

□ Waterfall Model Design:

The waterfall approach was the first SDLC Model to be used widely in Software Engineering to ensure the success of the project. In “The Waterfall” approach, the whole process of software development is divided into separate phases. In the Waterfall model, typically, the Outcome of one phase acts as the input for the next phase sequentially.

□ Iterative Waterfall Design:

Definition: The Iterative Waterfall Model is a variation of the traditional Waterfall model, which is a linear and sequential software development methodology. In the Iterative Waterfall Model, the development process is divided into small, manageable cycles, allowing for the revisiting and refinement of phases before progressing to the next stage. It combines the systematic structure of the Waterfall model with the flexibility of iterative development.

The sequential phases in Iterative Waterfall model are:

- **Requirement Gathering and Analysis:** All possible requirements of the system to be developed are captured in this phase and documented in a requirement specification doc.
- **System Design:** The requirement specifications from the first phase are studied in this phase and system design is prepared. System Design helps in specifying hardware and system requirements and also helps in defining overall system architecture.
- **Implementation:** With inputs from system design, the system is first developed in small programs called units, which are integrated in the next phase. Each unit is developed and tested for its functionality which is referred to as Unit Testing.
- **Integration and Testing:** All the units developed in the implementation phase are integrated into a system after testing each unit. Post integration the entire system is tested for any faults and failures.
- **Deployment of the system:** Once the functional and non-functional testing is done, the product is deployed in the customer environment or released into the market.
- **Maintenance:** Some issues come up in the client environment. To fix those issues patches are released. Also to enhance the product some better versions are released. Maintenance is done to deliver these changes in the customer environment.

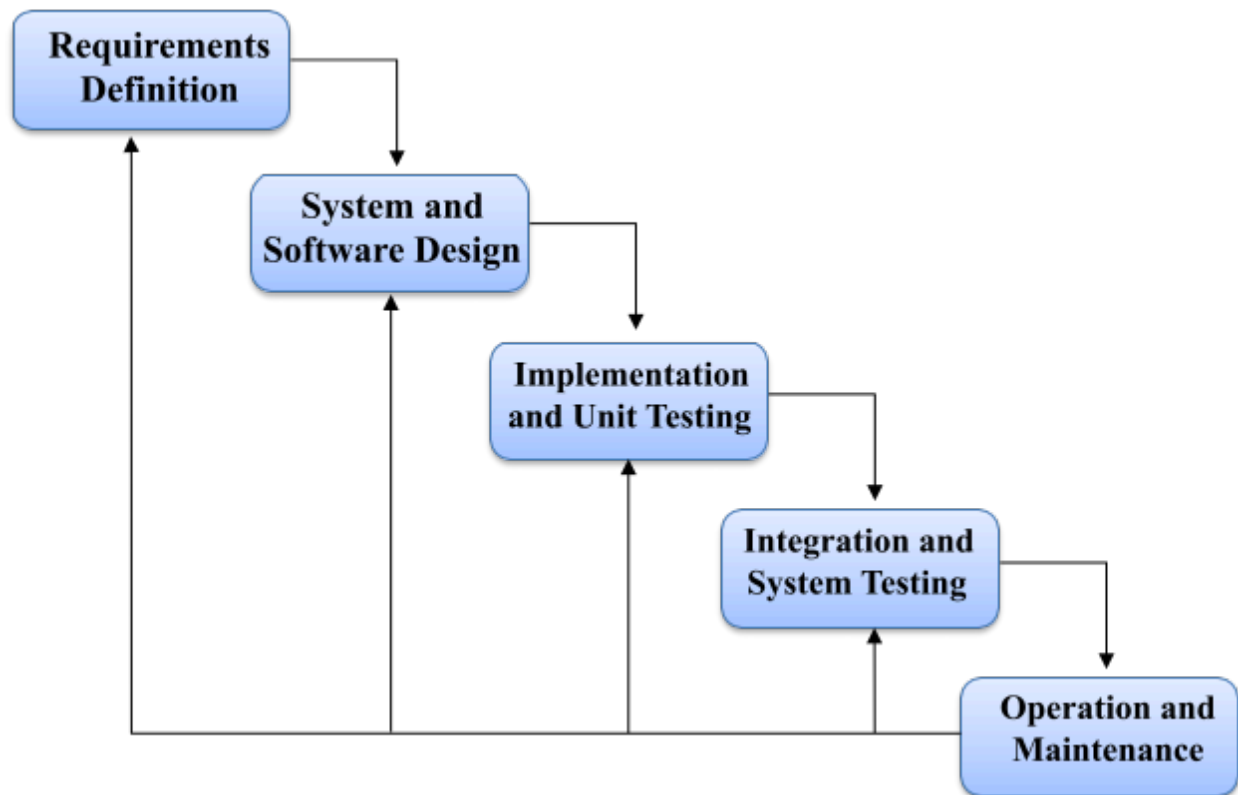
All these phases are cascaded to each other in progress and are seen as flowing steadily downwards (like a waterfall) through the phases. The next phase is started only after the defined set of goals are achieved for the previous phase and it is signed off, so the name “Iterative Waterfall Model”. In this model, phases do not overlap.

- **Advantages:**

- 1 . **Flexibility:** Iterations permit adjustments based on feedback.
- 2 . **Early Delivery:** Partial systems can be delivered incrementally.
- 3 . **Risk Management:** Identifying and addressing issues early in the process.

- **Disadvantages:**

1. **Increased Complexity:** The iterative nature can make the process more complex.
2. **Potential for Scope Creep:** Frequent iterations may lead to scope changes.
3. **Resource Intensive:** Continuous revisiting of phases may demand more resources.



- **Applications:**

The Iterative Waterfall Model is suitable for projects with evolving or unclear requirements. It is commonly used in software development projects where regular feedback and refinement are essential.

Additionally, it is applicable in scenarios where partial system delivery is beneficial, allowing stakeholders to assess progress and make adjustments.

3D .STUDY OF THE SYSTEM

Modules: The modules used in this software are as follows:

- **Register:**
 1. **College Register:** : Here, colleges will register to create their institutional profile and gain access to the communication and collaboration features.
- **Login:**
 1. **College Login:** Here, registered colleges will log in to access their dashboard, shared resources, events, forum, and other features. This includes authentication via email/password
- **Home(Dashboard):** This page serves as the main dashboard for the logged-in college, providing an overview of recent activities and quick access to core functionalities.
- **Password Management:**
 1. **Forgot Password:** Allows colleges to request a password reset link via their registered email.
 2. **Reset Password:** Enables colleges to set a new password using a unique token received via email.
- **College Profile, Resources & Event:**
 1. **My Profile, Edit Profile & Other College Profile:** Here, the logged-in college can view its own detailed profile, including contact information, address, description, and logo. Allows the logged-in college to update its profile details and logo. Enables colleges to view the public profiles of other registered institutions in the directory.
 2. **Upload Resources, View & Download Resources and Manage Own Resources:** Colleges can upload new documents categorized as Official Documents, Event Materials, Reports & Academic Content, or Administrative Documents. Colleges can browse, search, and securely download shared resources. Colleges can delete resources they have uploaded.
 3. **Create Event, View Events & Manage Own Events:** Colleges can create and post new inter-college events (workshops, seminars, fests, etc.). Colleges can browse upcoming events posted by other institutions. Colleges can delete events they have organized.
- **Forum/Discussions, Colleges Directory & In-App Notifications :**
 - **Create Post, View Posts & Manage Own Posts :** Colleges can initiate new discussion topics (posts) for the inter-college forum. Colleges can browse and participate in discussions, with the ability to see and mention other colleges within posts. Colleges can delete posts they have created.
 - **Colleges Directory:** This page provides a browsable and searchable directory of all other registered colleges, allowing for discovery and direct profile viewing.
 - **Notifications: In-App Notifications:** Provides real-time alerts to colleges about new resources, events, forum posts, and mentions, accessible via the Navbar.

3E.INPUT AND OUTPUT

The main inputs, outputs and the major function the details are:

□ **INPUT:**

1. **College Credentials:** Colleges provide email and password for secure login and registration.
2. **Profile Information:** Colleges input detailed information including name, address, contact details, description, website, and upload their institutional logo during registration or profile editing.
3. **Resource Data & Files:** Colleges submit resource titles, descriptions, categories, and upload relevant files (e.g., PDF, DOCX, images) for sharing.
4. **Event Details:** Colleges provide event titles, descriptions, types, locations, date/times, and optional contact/registration links for event creation.
5. **Forum Post Content:** Colleges input titles and content for forum discussions, including text-based mentions of other colleges.
6. **Search Queries:** Users type keywords into search bars to filter lists of colleges, resources, or events.
7. **Notification Actions:** Users interact by clicking notification items to view details or marking notifications as read.
8. **Password Reset Requests:** Colleges provide their email to initiate a password reset process.

□ **OUTPUT:**

1. **Authenticated Access:** Successful login provides access to the CollegeConnect platform and associated features.
2. **College Dashboard:** Displays an overview of platform activities, recent updates, and quick navigation links.
3. **College Profiles:** Presents detailed views of the logged-in college's own profile and public profiles of other institutions, including their logos.
4. **Resource Library:** Shows a browsable, searchable, and categorized list of shared documents with options for viewing (for supported file types) and downloading.
5. **Event Listings:** Displays a chronological list of upcoming events posted by various colleges.
6. **Forum Discussions:** Presents a list of forum posts, highlighting mentions of colleges.
7. **Notifications:** Provides real-time alerts via a Navbar icon and a detailed dropdown list of new activities (resources, events, forum posts, mentions).
8. **Confirmation & Error Toasts:** Displays immediate feedback to users for operations (e.g., successful upload, login errors, profile updates).
9. **Password Reset Links:** Sends unique, time-limited password reset links to the registered email addresses.

3F.SOFTWARE REQUIREMENT SPECIFICATIONS

This document outlines the essential functional and non-functional requirements for the CollegeConnect system, serving as a guide for development and testing.

Functional Requirements:

- A. College Authentication:** Secure registration and login (email/password, Google Sign-in), including robust password reset (forgot/reset password).
- B. Profile Management:** Allow colleges to create, view, and update their detailed profiles, including logo uploads.
- C. Resource Sharing:** Enable colleges to upload, browse, search, view, download, and manage categorized documents (e.g., Official, Event Materials).
- D. Event Management:** Facilitate creation, viewing, and deletion of inter-college events.
- E. Forum & Discussions:** Support creation, viewing, and deletion of discussion posts with college mentioning capabilities.
- F. Colleges Directory:** Provide a searchable directory to browse and view other college profiles.
- G. Notifications:** Deliver real-time in-app notifications for new content and activities, with read management.
- H. User Interface:** Offer a responsive, intuitive UI with consistent light/dark theme support.

Hardware Requirements:

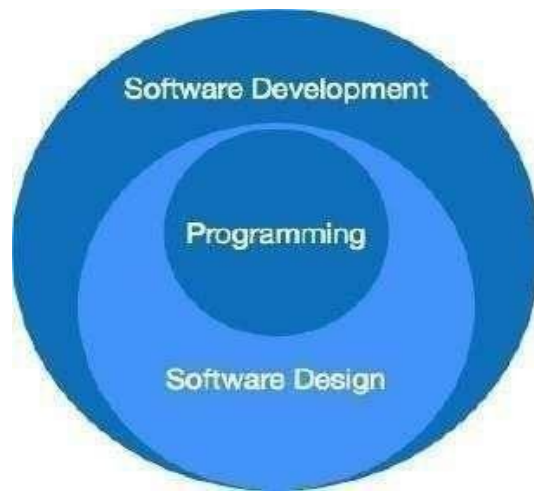
1. **Processor:** Intel Core i3 (or equivalent) or higher.
2. **RAM:** 8 GB or more.
3. **Storage:** SSD with at least 100 GB free space.

Software Requirements:

1. **Operating System:** Windows 10/11, macOS, or Linux.
2. **IDE:** Visual Studio Code.
3. **Database:** MongoDB.
4. **Runtime:** Node.js.
5. **Package Manager:** npm/Yarn.
6. **Browser:** Modern web browser.

3G.SOFTWARE ENGINEERING PARADIGM APPLIED

Software paradigms refer to the methods and steps, which are taken while designing the software. There are many methods proposed and are in work today, but we need to see where in software engineering these paradigms stand. These can be combined into various categories, though each of them is contained in one another.



The programming paradigm is a subset of Software design paradigm which is further a subset of the Software development paradigm.

There are two levels of reliability. The first is meeting the right requirements. A careful and thorough systems study is needed to satisfy this aspect of reliability. The second level of systems reliability involves the actual work delivered to the user. At this level, the system's reliability is interwoven with software engineering and development.

There are three approaches to reliability.

- 1. Error avoidance:** Prevents errors from occurring in software.
- 2. Error detection and correction:** In this approach, errors are recognized whenever they are encountered, and correcting the error by the effect of the error of the system does not fail.
- 3. Error tolerance:** In this approach, errors are recognized whenever they occur, but enables the system to keep running through degraded performance or Applying values that instruct the system to continue process.

4. SYSTEM DESIGN

4A. DATA FLOW DIAGRAM

A data flow diagram (DFD) is a graphical representation of the "flow" of data through an information system, modeling its process aspects. A DFD is often used as a preliminary step to create an overview of the system, which can later be elaborated.

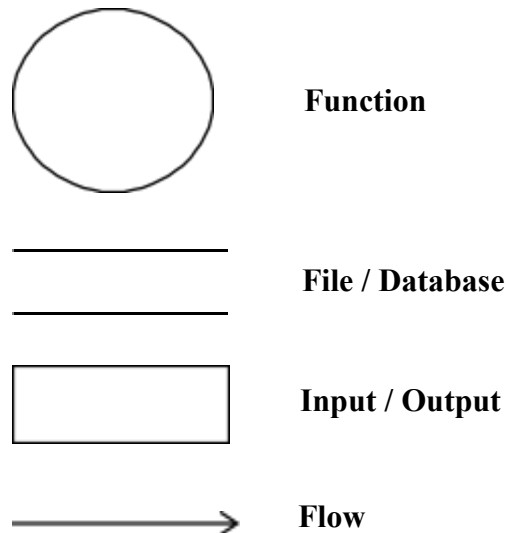
DFD can also be used for the visualization of data processing (structured design). A DFD shows what kind of information will be input to and output from the system, where the data will come from and go to, and where the data will be stored. It does not show information about the timing of the process or information about whether processes will operate in sequence or in parallel (which is shown on a flowchart).

This context-level DFD is next "exploded", to produce a Level 1 DFD that shows some of the detail of the system being modeled. The Level 1 DFD shows how the system is divided into sub- systems (processes), each of which deals with one or more of the data flows to or from an external agent, and which together provide all of the functionality of the system as a whole. It also identifies internal data stores that must be present for the system to do its job and shows the flow of data between the various parts of the system.

Data flow diagrams are one of the three essential perspectives of the structured- systems analysis and design method SSADM. The sponsor of a project and the end users will need to be briefed and consulted throughout all stages of a system's evolution. With a data flow diagram, users can visualize how the system will operate, what the system will accomplish, and how the system will be implemented. The old system's data-flow diagrams can be drawn up and compared with.

How any system is developed can be determined through a data flow diagram model. In the course of developing a set of leveled data flow diagrams, the analyst/designer is forced to address how the system may be decomposed into component sub-systems and to identify the transaction data in the data model. Data flow diagrams can be used in both the Analysis and Design phase of the SDLC. There are different notations to draw data flow diagrams. Defining different visual representations for processes, data stores, data flow, and external entities.

DFD Notation:



DFD Example:



Steps to Construct Data Flow Diagram:

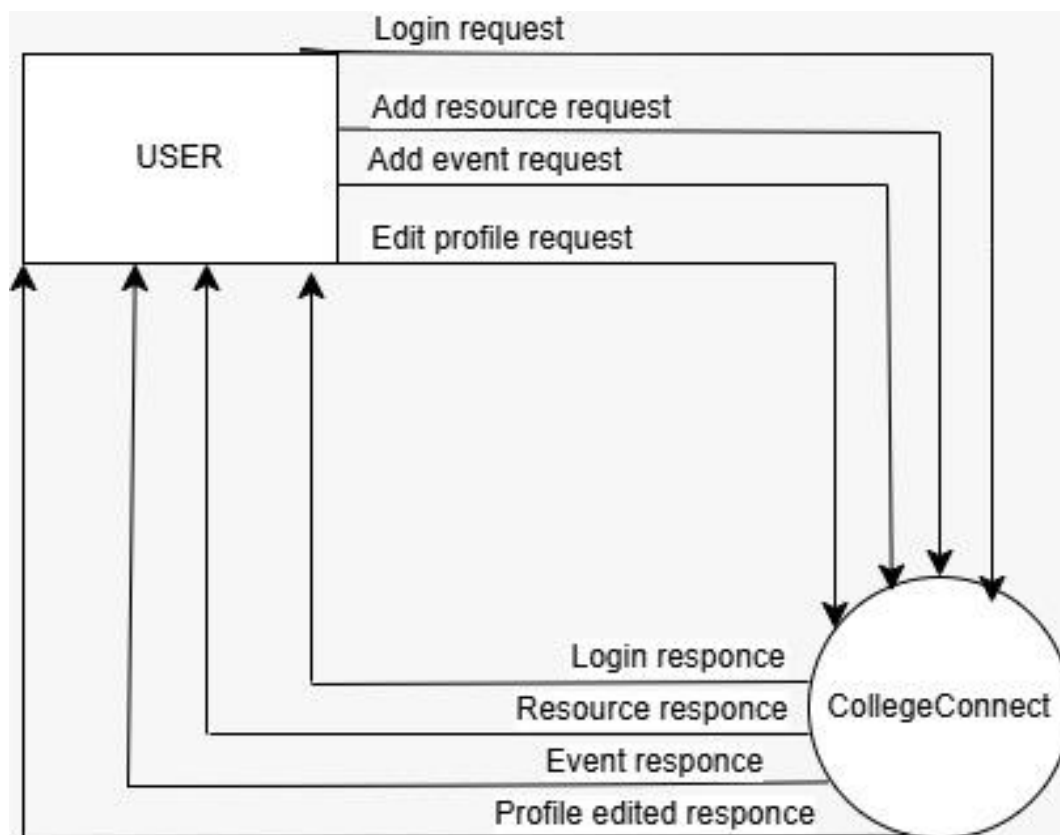
Four Steps are generally used to construct a DFD.

- Process should be named and referred for easy reference. Each name should be representative of the reference.
- The destination of flow is from top to bottom and from left to right.
- When a process is distributed into lower-level details they are numbered.
- The names of data stores, sources, and destinations are written in capital letters.

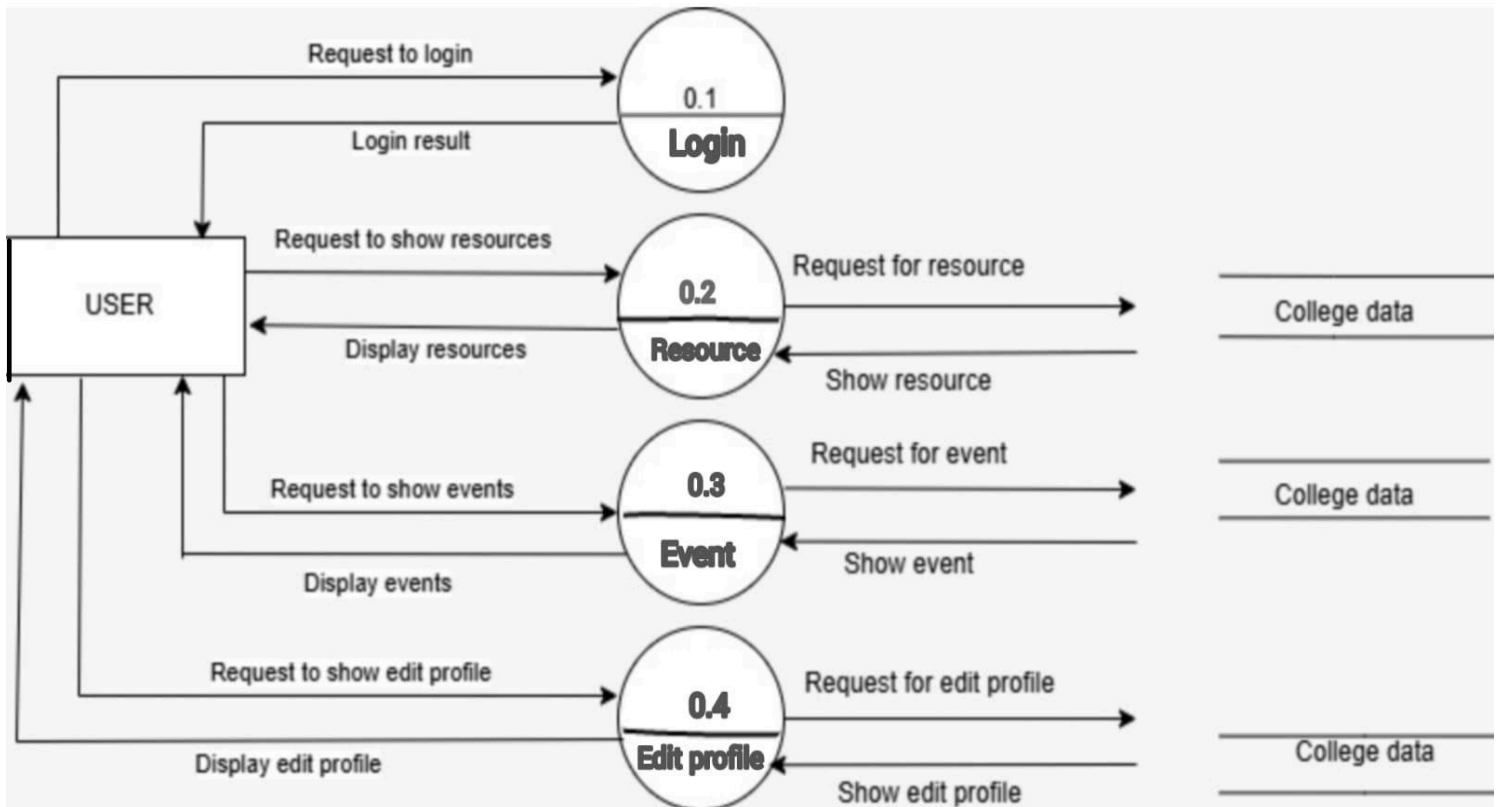
Rules for constructing a Data Flow Diagram:

- Arrows should not cross each other.
- Squares, Circles, and Files must bear a name.
- Decomposed data flow squares and circles can have the same names.
- Draw all data flow around the outside of the diagram.

- **LEVEL 0 DFD :**



- **LEVEL 1 DFD:**



4B.SEQUENCE DIAGRAM

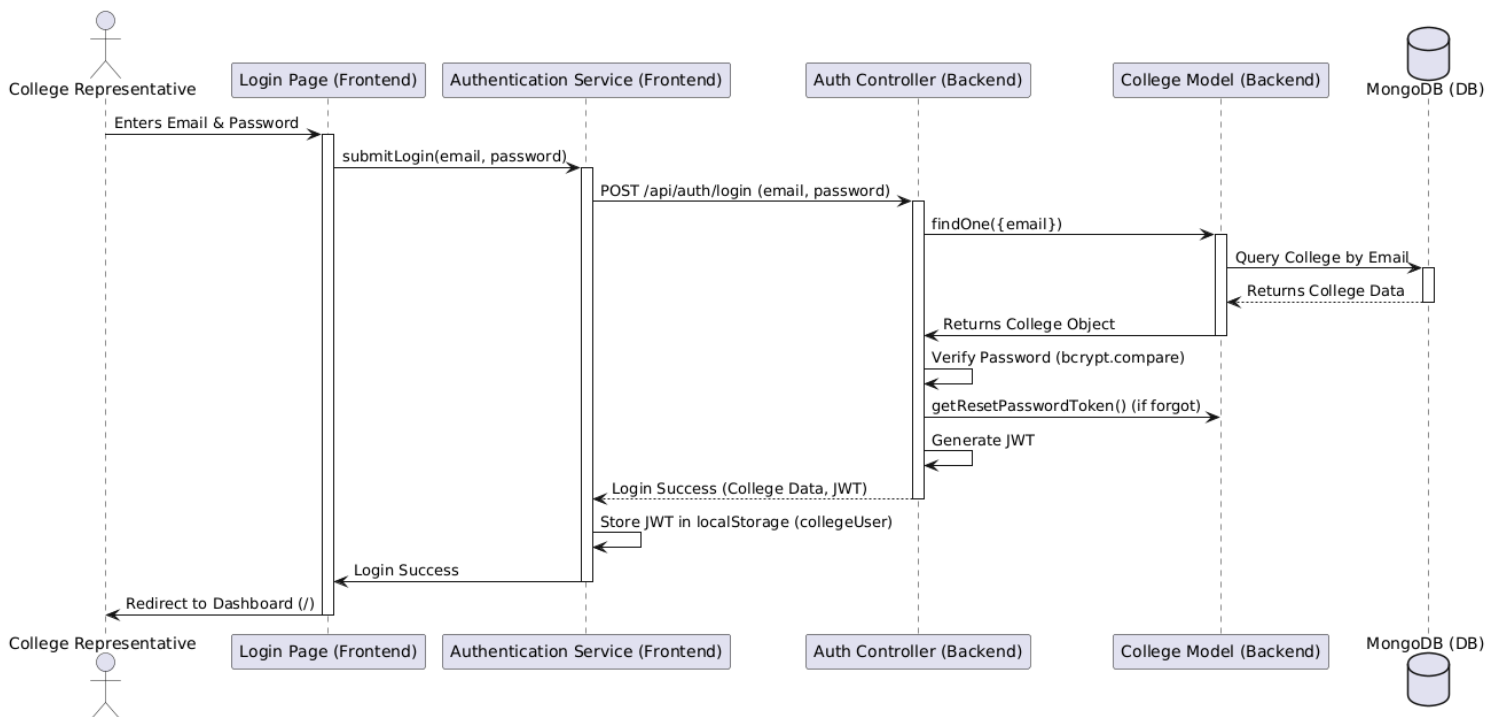
A Sequence diagram is an interaction diagram that shows how processes operate with one another and what is their order. It is a construct of a Message Sequence Chart. A sequence diagram shows object interactions arranged in a time sequence. It depicts the objects and classes involved in the scenario and the sequence of messages exchanged between the objects needed to carry out the functionality of the scenario. Sequence diagrams are typically associated with use case realizations in the Logical View of the system under development.

Sequence diagrams are sometimes called event diagrams or event scenarios.

A sequence diagram shows, as parallel vertical lines (lifelines), different processes or objects that live simultaneously, and, as horizontal arrows, the messages exchanged between them, in the order in which they occur. This allows the specification of simple runtime scenarios in a graphical manner.

A sequence diagram is the most common kind of interaction diagram, which focuses on the message interchange between several life lines .A sequence diagram describes an interaction by focusing on the sequence of messages that are exchanged, along with their corresponding occurrence specifications on the lifelines.

The following nodes and edges are typically drawn in a UML sequence diagram: lifeline, execution specification, message, fragment, interaction, state invariant, continuation, and destruction occurrence.



A Use case diagram at its simplest is a representation of a user's interaction with the system that shows the relationship between the user and the different use cases in which the user is involved. A use case diagram can identify the different types of users of a system and the different use cases and will often be accompanied by other types of diagrams as well.

So only static behavior is not sufficient to model a system rather dynamic behavior is more important than static behavior. In UML there are five diagrams available to model dynamic nature and a use case diagram is one of them. Now as we have to discuss that the use case diagram is dynamic in nature there should be some internal or external factors for making the interaction.

These internal and external agents are known as actors. So, use case diagrams consist of actors, use cases, and their relationships. The diagram is used to model the system/subsystem of an application. A single-use case diagram captures a particular functionality of a system.

So, to model the entire system numbers of use case diagrams are used. The purpose of a use case diagram is to capture the dynamic aspect of a system. But this definition is too generic to describe the purpose.

Because the other four diagrams (activity, sequence, collaboration, and State chart) are also having the same purpose. So, we will look into some specific purpose that will distinguish it from the other four diagrams.

Use case diagrams are used to gather the requirements of a system including internal and external influences. These requirements are mostly design requirements. So, when a system is analyzed to gather its functionalities use cases are prepared and actors are identified.

Now when the initial task is complete use case diagrams are modeled to present the outside view. So, in brief, the purposes of use case diagrams can be as follows:

- ☐ Used to gather requirements of a system.
- ☐ Used to get an outside view of a system.
- ☐ Identify external and internal factors influencing the system.

How to draw Use Case Diagram?

Use case diagrams are considered for high level requirement analysis of a system. So, when the requirements of a system are analyzed, the functionalities are captured in use cases.

So, we can say that uses cases are nothing but the system functionalities written in an organized manner. Now the second things which are relevant to the use cases are the actors. Actors can be defined as something that interacts with the system.

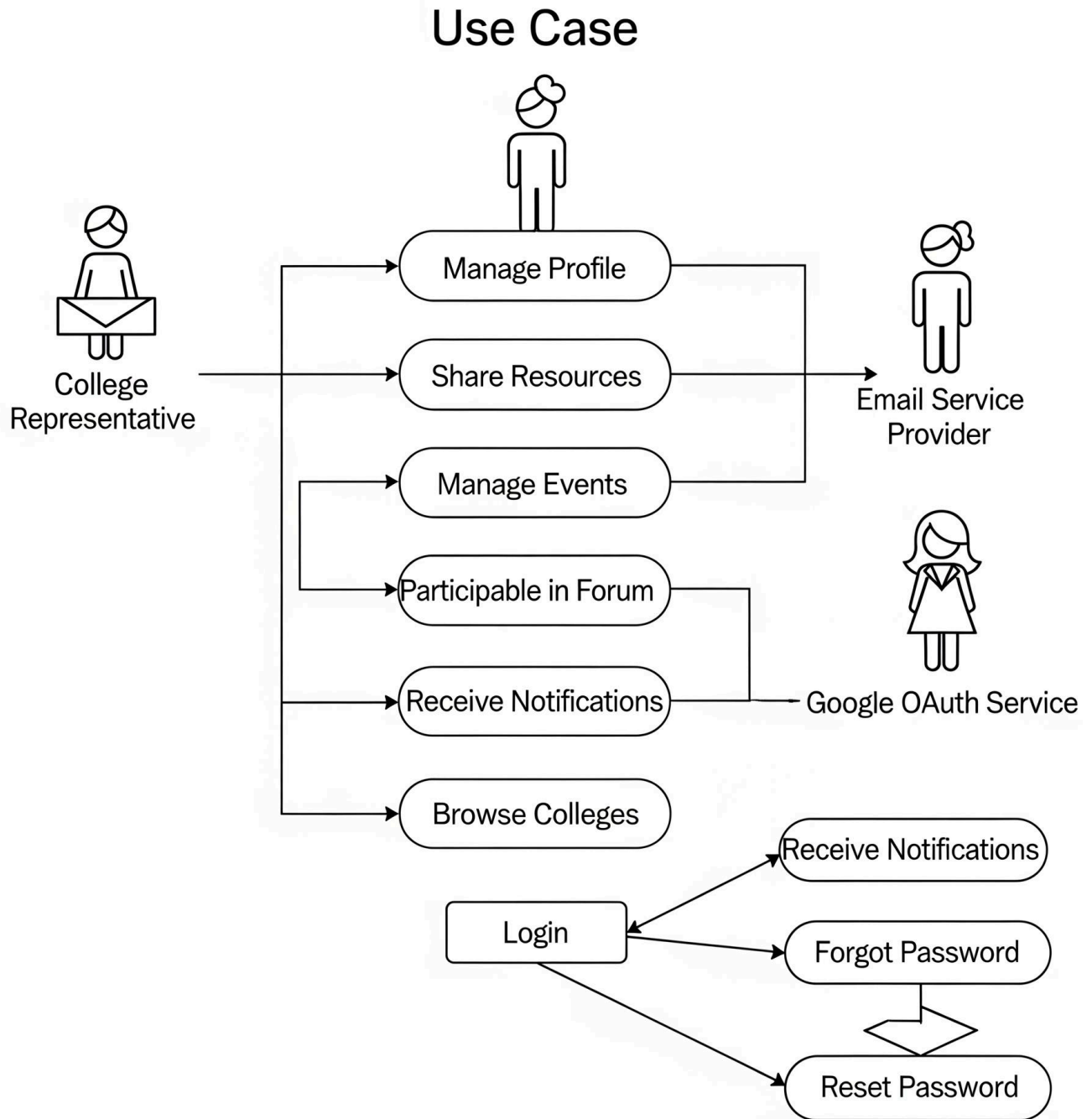
The actors can be human user, some internal applications or may be some external applications. So, in a brief when we are planning to draw use case diagram, we should have the following items identified.

- ☐ Functionalities to be represented as a use case
- ☐ Actors
- ☐ Relationships among the use cases and actors.

Use case diagrams are drawn to capture the functional requirements of a system. So, after identifying the above items we have to follow the following guidelines to draw an efficient use case diagram.

- ☐ The name of a use case is very important. So, the name should be chosen in such a way so that it can identify the functionalities performed.
- ☐ Give a suitable name for actors.
- ☐ Show relationships and dependencies clearly in the diagram.
- ☐ Do not try to include all types of relationships. Because the main purpose of the diagram is to identify requirements.
- ☐ Use note whenever required to clarify some important point

- USE CASE DIAGRAM:

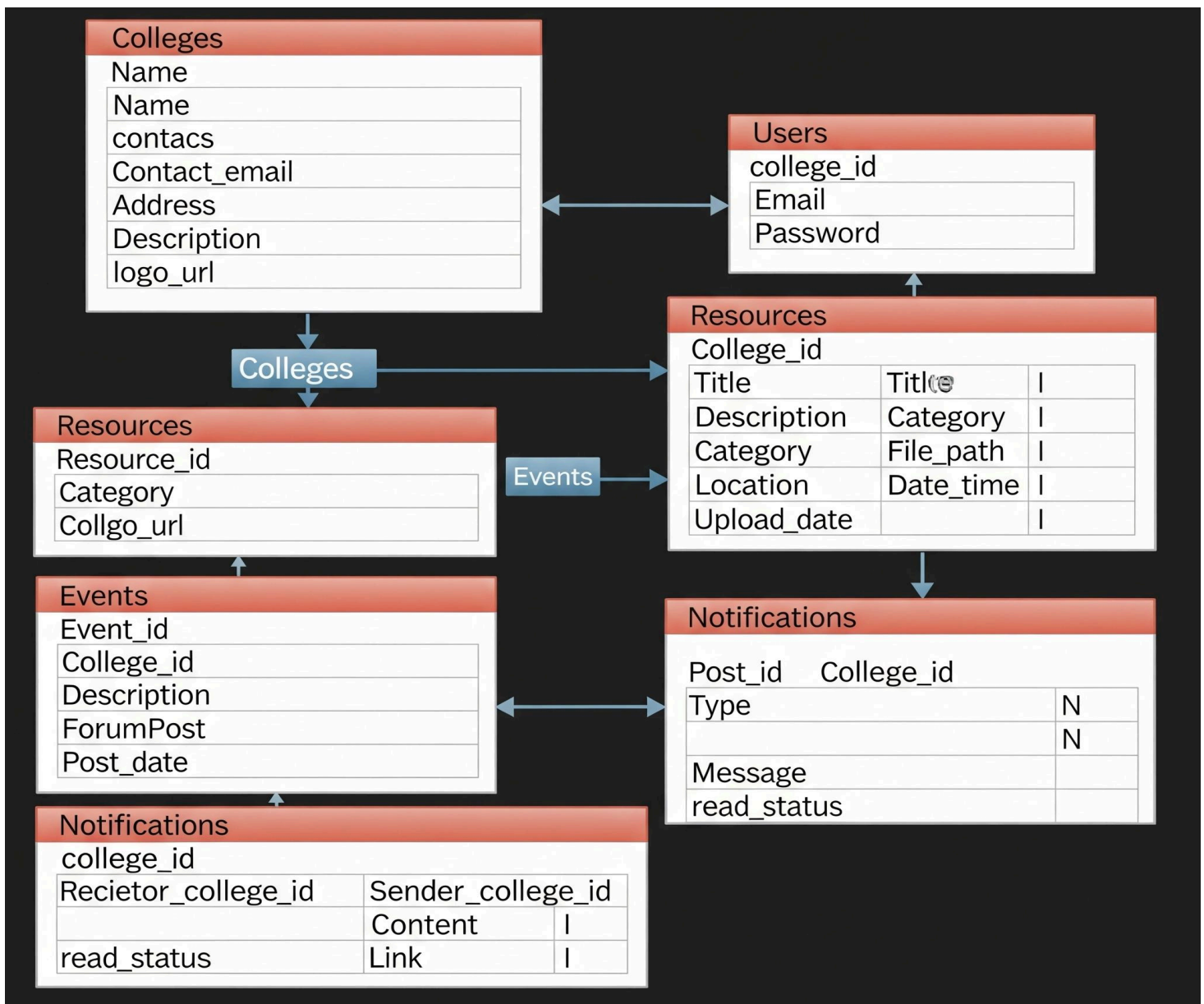


4D.SCHEMA DIAGRAM

The schema is an abstract structure or outline representing the logical view of the database as a whole. Defining categories of data and relationships between those categories, database schema design makes data much easier to retrieve, consume, manipulate, and interpret.

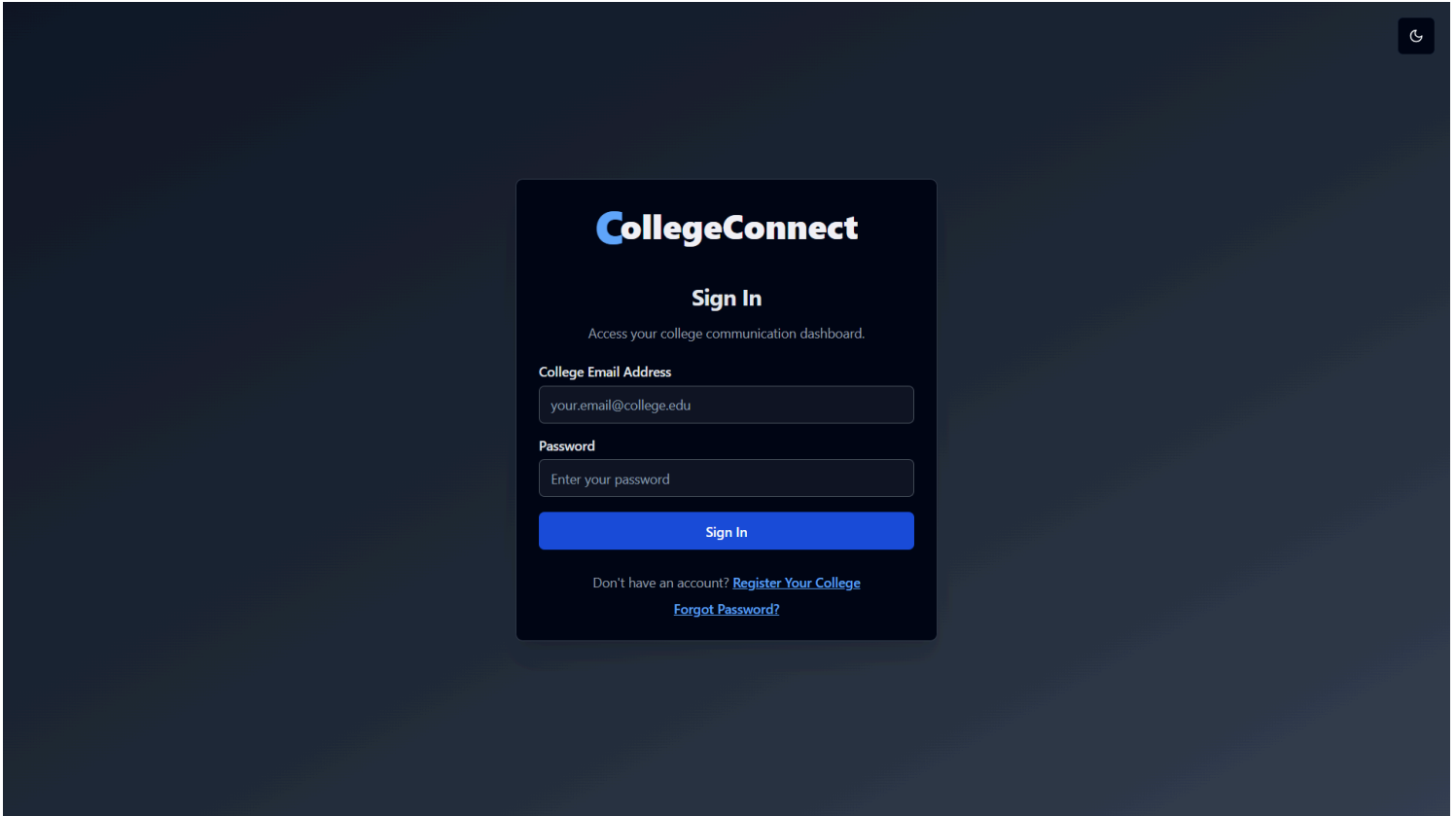
DB schema design organizes data into separate entities, determines how to create relationships between organized entities, and influences the applications of constraints on data. Designers create database schema to give other database users, such as programmers and analysts, a logical understanding of data.

- **SCHEMA DESIGN:**



5. UI SNAPSHOT

5A. FRONTEND



Login page:

return (

```
<div className="min-h-screen flex items-center justify-center bg-gradient-to-br from-blue-200 via-purple-200 to-blue-300 dark:from-gray-900 dark:via-gray-800 dark:to-gray-700 p-4 relative">
```

```
  { /* ModeToggle restored */ }
```

```
  <div className="absolute top-4 right-4 z-10">
```

```
    <ModeToggle />
```

```
  </div>
```

```
  <Card className="w-full max-w-md mx-auto shadow-xl rounded-lg border border-gray-100 dark:border-gray-700">
```

```
    <CardHeader className="space-y-3 text-center">
```

```
      { /* Logo with a styled "C" and CollegeConnect text */ }
```

```
      <div className="flex items-center justify-center mb-6">
```

```
        <span className="text-blue-600 dark:text-blue-400 text-5xl font-extrabold -mr-1">C</span>
```

```
        <h1 className="text-4xl font-extrabold text-gray-800 dark:text-gray-100">CollegeConnect</h1>
```

```
      </div>
```

```
      { /* Sign In title and description */ }
```

```
      <CardTitle className="text-2xl font-bold text-gray-700 dark:text-gray-200 mt-2">Sign In</CardTitle>
```

```

    <CardDescription className="text-gray-600 dark:text-gray-400">Access your college communication
    dashboard.</CardDescription>
  </CardHeader>
  <CardContent>
    <form onSubmit={handleSubmit} className="space-y-4">
      <div className="grid gap-2">
        <Label htmlFor="email" className="text-gray-700 dark:text-gray-200">College Email Address</Label>
        <Input
          id="email"
          type="email"
          placeholder="your.email@college.edu"
          value={email}
          onChange={(e) => setEmail(e.target.value)}
          required
          className="bg-white dark:bg-gray-900 border-gray-300 dark:border-gray-600 focus:ring-blue-500
focus:border-blue-500 rounded-md py-2 px-3"
        />
      </div>
      <div className="grid gap-2">
        <Label htmlFor="password" className="text-gray-700 dark:text-gray-200" >Password</Label>
        <Input
          id="password"
          type="password"
          placeholder="Enter your password"
          value={password}
          onChange={(e) => setPassword(e.target.value)}
          required
          className="bg-white dark:bg-gray-900 border-gray-300 dark:border-gray-600 focus:ring-blue-500
focus:border-blue-500 rounded-md py-2 px-3"
        />
      </div>
      <Button
        type="submit"
        className="w-full bg-blue-600 hover:bg-blue-700 dark:bg-blue-700 dark:hover:bg-blue-600 text-white
font-semibold py-2 rounded-md transition-colors duration-200 ease-in-out"
        disabled={isLoading}
      >
        {isLoading ? 'Signing In...' : 'Sign In'}
      </Button>
    </form>
    <div className="mt-6 text-center text-sm text-gray-600 dark:text-gray-400">
      <p className="mb-2">Don't have an account?{" "}</p>
      <Link to="/register" className="text-blue-600 hover:text-blue-700 dark:text-blue-400
dark:hover:text-blue-300 font-medium transition-colors duration-200 ease-in-out underline">
        Register Your College
      </Link>
    </div>
  </CardContent>
</Card>

```



```

    </p>
    <p>
      <Link to="/forgot-password" className="text-blue-600 hover:text-blue-700 dark:text-blue-400
dark:hover:text-blue-300 font-medium transition-colors duration-200 ease-in-out underline">
        Forgot Password?
      </Link>
    </p>
  </div>
</CardContent>
</Card>
<Toaster />
</div>
);
};

```

```
export default LoginPage;
```

Register Page:

```

const isValidForm = () => {
  return (
    name &&
    email &&

```

```

password &&
confirmPassword &&
address &&
city &&
state &&
country &&
description &&
passwordsMatch &&
!emailError
);
};

return (
  <div className="min-h-screen flex items-center justify-center bg-gradient-to-br from-blue-200 via-purple-200 to-blue-300 dark:from-gray-900 dark:via-gray-800 dark:to-gray-700 p-4 relative">
    { /* ModeToggle restored */ }
    <div className="absolute top-4 right-4 z-10">
      <ModeToggle />
    </div>
    <Card className="w-full max-w-2xl mx-auto shadow-xl rounded-lg border border-gray-100 dark:border-gray-700">
      <CardHeader className="space-y-3 text-center">
        { /* CollegeConnect logo text */ }
        <div className="flex items-center justify-center mb-6">
          <span className="text-blue-600 dark:text-blue-400 text-5xl font-extrabold -mr-1">C</span>
          <h1 className="text-4xl font-extrabold text-gray-800 dark:text-gray-100">ollegeConnect</h1>
        </div>
        { /* Registration title and description */ }
        <CardTitle className="text-2xl font-bold text-gray-700 dark:text-gray-200 mt-2">Register Your College</CardTitle>
        <CardDescription className="text-gray-600 dark:text-gray-400">Enter your college details to create an account</CardDescription>
      </CardHeader>
      <CardContent>
        <form onSubmit={handleSubmit} className="grid grid-cols-1 md:grid-cols-2 gap-x-6 gap-y-4">
          { /* Basic Details */ }
          <div className="grid gap-2">
            <Label htmlFor="name" className="text-gray-700 dark:text-gray-200">College Name</Label>
            <Input
              id="name"
              type="text"
              placeholder="Enter your college name"
              value={name}
              onChange={(e) => setName(e.target.value)}
              required
              className="bg-white dark:bg-gray-900 border-gray-300 dark:border-gray-600 focus:ring-blue-500

```

```

focus:border-blue-500 rounded-md py-2 px-3"
  />
</div>
<div className="grid gap-2">
  <Label htmlFor="email" className="text-gray-700 dark:text-gray-200">Contact Email</Label>
  <Input
    id="email"
    type="email"
    placeholder="name@example.com"
    value={email}
    onChange={(e) => {
      setEmail(e.target.value);
      validateEmail(e.target.value);
    }}
    onBlur={(e) => validateEmail(e.target.value)}
    required
    className="bg-white dark:bg-gray-900 border-gray-300 dark:border-gray-600 focus:ring-blue-500
focus:border-blue-500 rounded-md py-2 px-3"
  />
  {emailError && (
    <p className="text-red-500 text-sm mt-1">{emailError}</p>
  )}
</div>

{/* Password Fields */}
<div className="grid gap-2">
  <Label htmlFor="password" className="text-gray-700 dark:text-gray-200">Password</Label>
  <Input
    id="password"
    type="password"
    placeholder="Enter your password"
    value={password}
    onChange={(e) => setPassword(e.target.value)}
    required
    className="bg-white dark:bg-gray-900 border-gray-300 dark:border-gray-600 focus:ring-blue-500
focus:border-blue-500 rounded-md py-2 px-3"
  />
</div>
<div className="grid gap-2">
  <Label htmlFor="confirm-password" className="text-gray-700 dark:text-gray-200">Confirm
Password</Label>
  <Input
    id="confirm-password"
    type="password"
    placeholder="Confirm your password"
    value={confirmPassword}

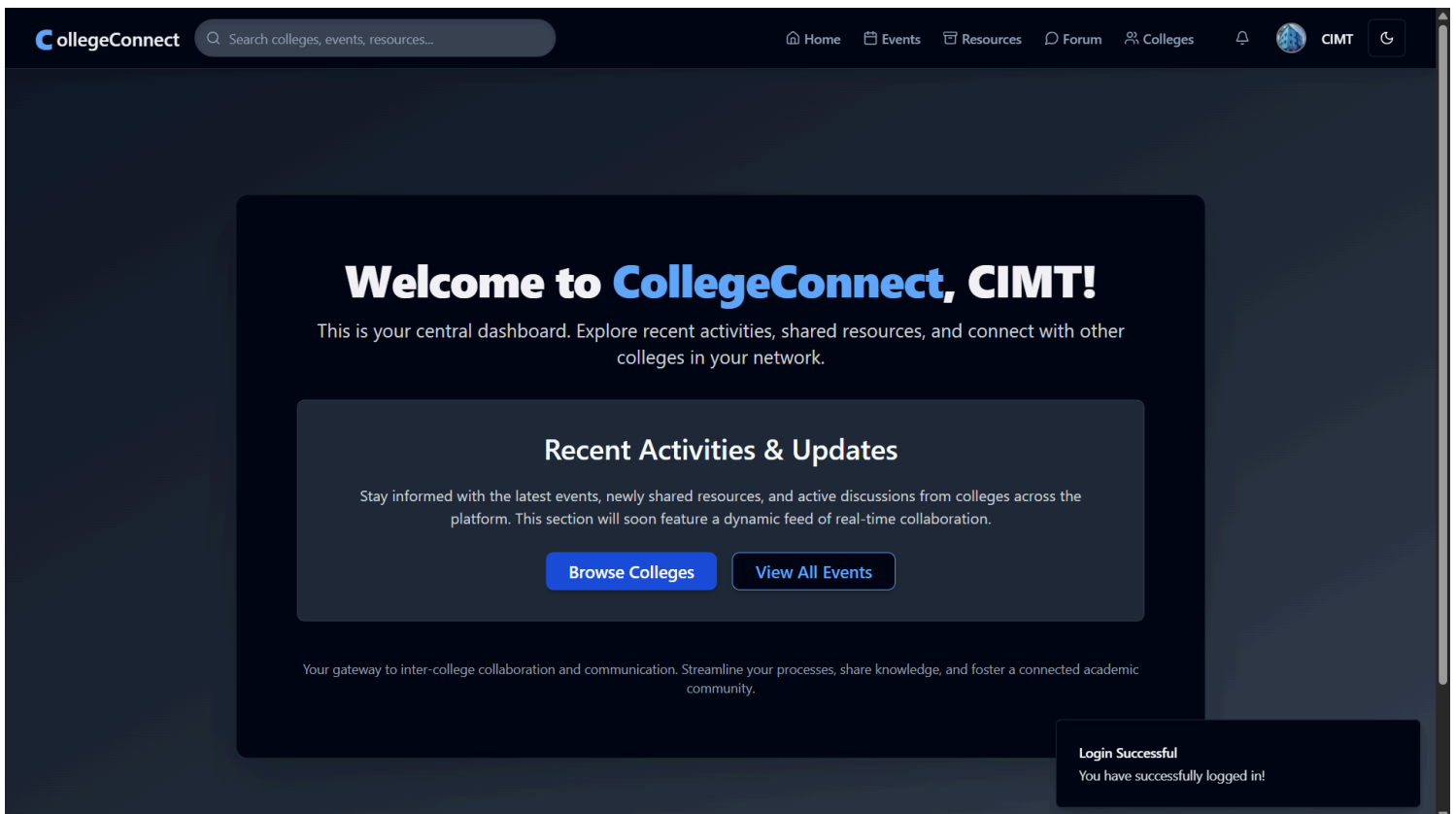
```

```

      onChange={(e) => setConfirmPassword(e.target.value)}
      required
      className="bg-white dark:bg-gray-900 border-gray-300 dark:border-gray-600 focus:ring-blue-500
focus:border-blue-500 rounded-md py-2 px-3"
    />
    {isLoading ? 'Registering...' : 'Register College'}
  </Button>
</div>
</form>
<div className="mt-6 text-center text-sm text-gray-600 dark:text-gray-400">
  Already have an account?{" "}
  <Link to="/login" className="text-blue-600 hover:text-blue-700 dark:text-blue-400
dark:hover:text-blue-300 font-medium transition-colors duration-200 ease-in-out underline">
    Login
  </Link>
</div>
</CardContent>
</Card>
<Toaster />
</div>
);
};

export default RegisterPage;

```



Home Page:

```
return (  
  // Applied consistent gradient background and increased overall padding  
  <div className="min-h-screen flex flex-col items-center justify-center bg-gradient-to-br from-blue-200  
via-purple-200 to-blue-300 dark:from-gray-900 dark:via-gray-800 dark:to-gray-700 p-6 md:p-16 lg:p-20 xl:p-24">  
    { /* Increased max-w to fill more horizontal space, and adjusted inner padding */}  
    <div className="text-center bg-card text-card-foreground p-8 md:p-12 lg:p-16 rounded-xl shadow-2xl  
max-w-5xl w-full transform transition-all duration-300 hover:scale-[1.005] hover:shadow-3xl">  
      <h1 className="text-4xl md:text-5xl font-extrabold text-gray-800 dark:text-gray-100 mb-4 tracking-tight">  
        Welcome to <span className="text-blue-600 dark:text-blue-400">CollegeConnect</span>, {collegeUser ?  
collegeUser.name : 'College'}!  
      </h1>  
      <p className="text-lg md:text-xl mb-8 text-gray-600 dark:text-gray-300 leading-relaxed">  
        This is your central dashboard. Explore recent activities, shared resources, and connect with other colleges in  
your network.  
      </p>  
  
      { /* Enhanced Placeholder for Activity Feed */}  
      <div className="bg-white dark:bg-gray-800 text-foreground p-8 rounded-lg shadow-lg border  
border-gray-200 dark:border-gray-700 mt-8">  
        <h2 className="text-2xl md:text-3xl font-semibold text-gray-800 dark:text-gray-50 mb-5">Recent Activities &  
Updates</h2>  
        <p className="text-base text-gray-700 dark:text-gray-300 leading-normal">  
          Stay informed with the latest events, newly shared resources, and active discussions from colleges across the  
platform. This section will soon feature a dynamic feed of real-time collaboration.  
        </p>  
        <div className="mt-6 flex flex-col sm:flex-row justify-center gap-4">  
          <Button  
            onClick={() => navigate('/colleges')}  
            className="px-6 py-3 text-lg bg-blue-600 hover:bg-blue-700 dark:bg-blue-700 dark:hover:bg-blue-600  
text-white font-semibold rounded-lg shadow-md transition-colors duration-300 ease-in-out transform  
hover:scale-105"  
          >  
            Browse Colleges  
          </Button>  
          <Button  
            variant="outline"  
            onClick={() => navigate('/events')}  
            className="px-6 py-3 text-lg border-blue-600 text-blue-600 hover:bg-blue-50 dark:border-blue-400  
dark:text-blue-400 dark:hover:bg-gray-700 font-semibold rounded-lg shadow-md transition-colors duration-300  
ease-in-out transform hover:scale-105"  
          >  
            View All Events  
          </Button>  
        </div>  
      </div>  
    </div>  
  )  
)
```

```

    <p className="mt-10 text-sm text-gray-500 dark:text-gray-400">
      Your gateway to inter-college collaboration and communication. Streamline your processes, share
      knowledge, and foster a connected academic community.
    </p>
  </div>
</div>
);
};

export default HomePage;

```

Event Page:

```

return (
  <div className="container mx-auto py-8 px-4 md:px-6 lg:px-8">
    <h1 className="text-4xl font-extrabold text-gray-800 dark:text-gray-100 mb-8 text-center">
      College <span className="text-blue-600 dark:text-blue-400">Events</span>
    </h1>

    { /* Create New Event Section */ }
    <Card className="mb-8 shadow-lg border border-gray-100 dark:border-gray-700">
      <CardHeader>
        <CardTitle className="text-2xl font-bold flex items-center gap-2">
          <CalendarPlusIcon className="h-6 w-6 text-blue-600 dark:text-blue-400" /> Create New Event

```

```

</CardTitle>
<CardDescription>Announce your college's upcoming events to the network.</CardDescription>
</CardHeader>
<CardContent>
  <form onSubmit={handleCreateEvent} className="grid grid-cols-1 md:grid-cols-2 gap-4">
    <div className="grid gap-2">
      <Label htmlFor="title">Event Title</Label>
      <Input id="title" type="text" value={title} onChange={(e) => setTitle(e.target.value)} required />
    </div>
    <div className="grid gap-2">
      <Label htmlFor="eventType">Event Type</Label>
      <Select value={eventType} onChange={setEventType} required>
        <SelectTrigger className="w-full">
          <SelectValue placeholder="Select event type" />
        </SelectTrigger>
        <SelectContent>
          {eventTypes.map(type => (
            <SelectItem key={type} value={type}>{type}</SelectItem>
          ))}
        </SelectContent>
      </Select>
    </div>
    <div className="grid gap-2 col-span-full">
      <Label htmlFor="description">Description</Label>
      <Textarea id="description" placeholder="Provide a detailed description of the event..."
value={description} onChange={(e) => setDescription(e.target.value)} required rows={3} />
    </div>
    <div className="grid gap-2">
      <Label htmlFor="location" className="flex items-center gap-2">
        <MapPinIcon className="h-4 w-4 text-muted-foreground" /> Location (Physical Address or Link):
      </Label>
      <Input id="location" type="text" value={location} onChange={(e) => setLocation(e.target.value)}
required />
    </div>
    <div className="grid gap-2">
      <Label htmlFor="dateTime" className="flex items-center gap-2">
        <CalendarIcon className="h-4 w-4 text-muted-foreground" /> Date & Time:
      </Label>
      { /* Using type="datetime-local" for convenient date/time input */ }
      <Input id="dateTime" type="datetime-local" value={dateTime} onChange={(e) =>
setDateTime(e.target.value)} required />
    </div>
    <div className="grid gap-2">
      <Label htmlFor="contactEmail" className="flex items-center gap-2">
        <MailIcon className="h-4 w-4 text-muted-foreground" /> Contact Email (Optional):
      </Label>

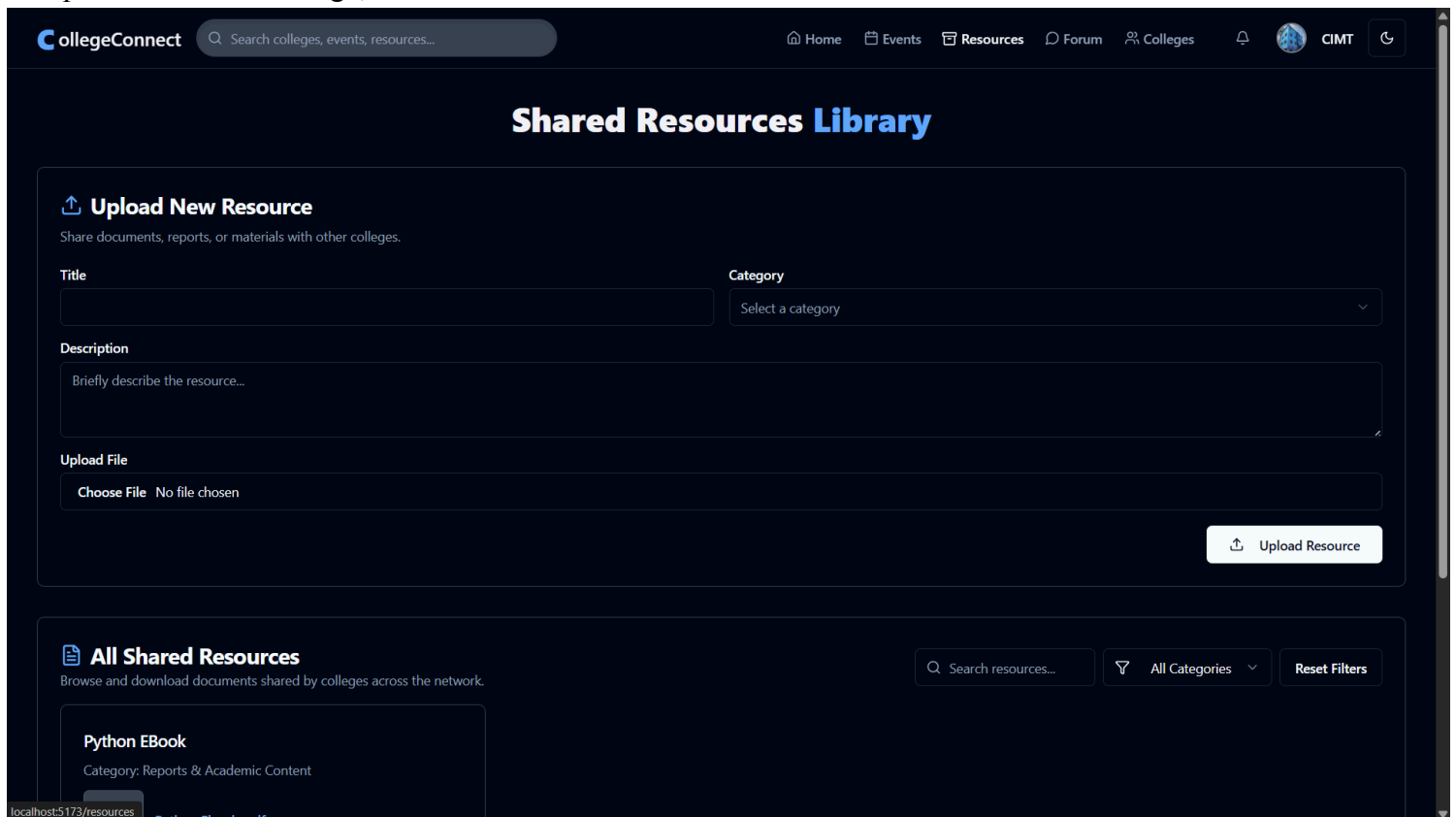
```

```

<Button
  variant="destructive"
  size="sm"
  onClick={() => handleDeleteEvent(event._id, event.title)}
  className="flex items-center gap-1"
>
  <Trash2Icon className="h-4 w-4" /> Delete
</Button>
</div>
  )}
</div>
</Card>
  )}
</div>
  )}
</CardContent>
</Card>
<Toaster />
</div>
);
};

```

export default EventsPage;



Resource Page:

```
return (
  <div className="container mx-auto py-8 px-4 md:px-6 lg:px-8">
    <h1 className="text-4xl font-extrabold text-gray-800 dark:text-gray-100 mb-8 text-center">
      Shared Resources <span className="text-blue-600 dark:text-blue-400">Library</span>
    </h1>

    { /* Upload New Resource Section */ }
    <Card className="mb-8 shadow-lg border border-gray-100 dark:border-gray-700">
      <CardHeader>
        <CardTitle className="text-2xl font-bold flex items-center gap-2">
          <UploadIcon className="h-6 w-6 text-blue-600 dark:text-blue-400" /> Upload New Resource
        </CardTitle>
        <CardDescription>Share documents, reports, or materials with other colleges.</CardDescription>
      </CardHeader>
      <CardContent>
        <form onSubmit={handleUploadSubmit} className="grid grid-cols-1 md:grid-cols-2 gap-4">
          <div className="grid gap-2">
            <Label htmlFor="title">Title</Label>
            <Input id="title" type="text" value={title} onChange={(e) => setTitle(e.target.value)} required />
          </div>
          <div className="grid gap-2">
            <Label htmlFor="category">Category</Label>
            <Select value={category} onChange={setCategory} required>
              <SelectTrigger className="w-full">
                <SelectValue placeholder="Select a category" />
              </SelectTrigger>
              <SelectContent>
                {resourceCategories.map(cat => (
                  <SelectItem key={cat} value={cat}>{cat}</SelectItem>
                ))}
              </SelectContent>
            </Select>
          </div>
          <div className="grid gap-2 col-span-full">
            <Label htmlFor="description">Description</Label>
            <Textarea id="description" placeholder="Briefly describe the resource..." value={description}
              onChange={(e) => setDescription(e.target.value)} required rows={3} />
          </div>
          <div className="grid gap-2 col-span-full">
            <Label htmlFor="resourceFile">Upload File</Label>
            <Input id="resourceFile" type="file" onChange={(e) => setResourceFile(e.target.files[0])} required />
            {resourceFile && <p className="text-sm text-gray-500">Selected: {resourceFile.name}</p>}
          </div>
          <div className="col-span-full flex justify-end">
            <Button type="submit" disabled={isUploading} className="px-6 py-2">

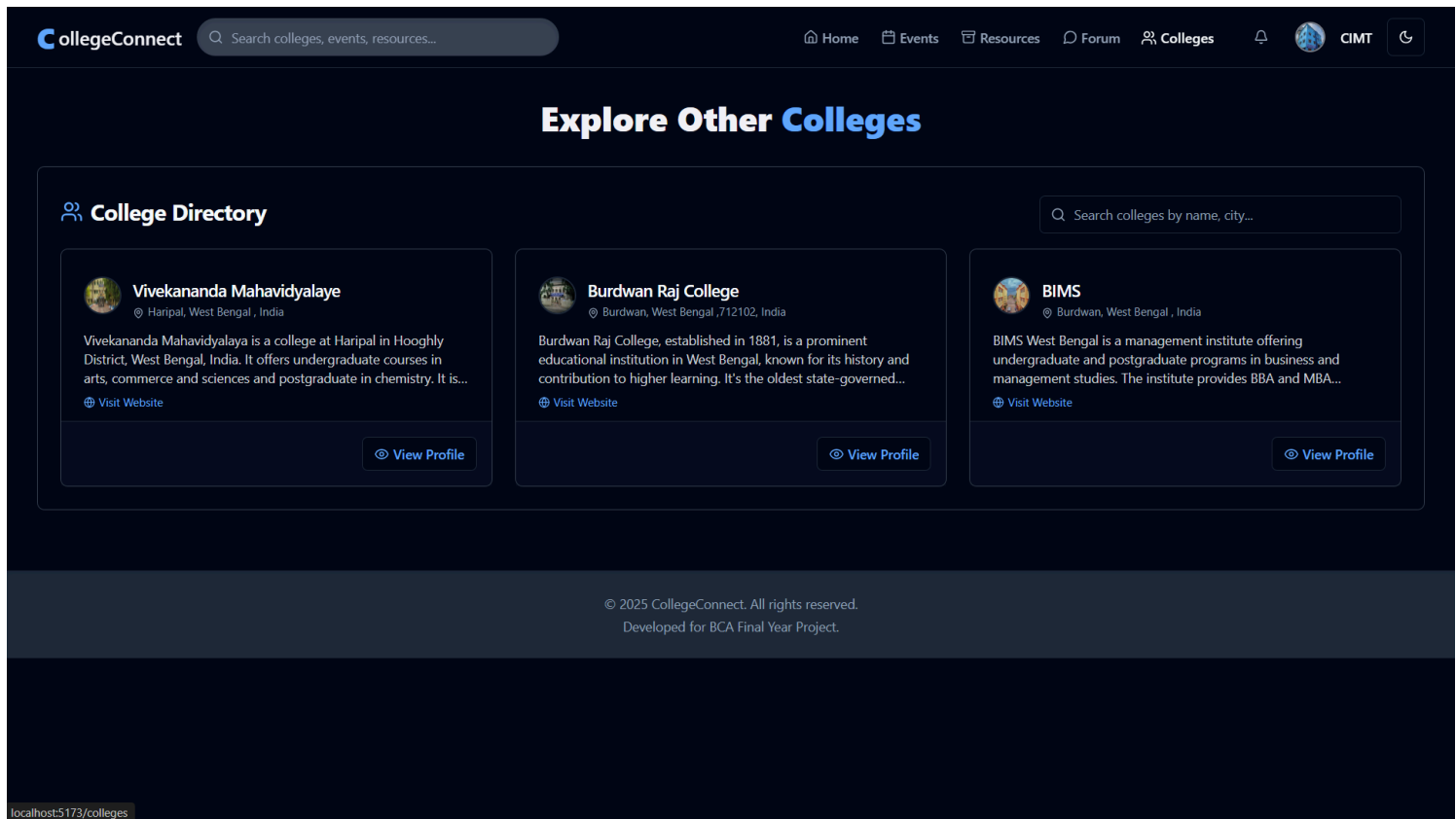
```

```

        {isUploading ? <Loader2 className="mr-2 h-4 w-4 animate-spin" /> : <UploadIcon className="mr-2
h-4 w-4" />}
        {isUploading ? 'Uploading...' : 'Upload Resource'}
      </Button>
    </div>
  </form>
</CardContent>
</Card>

{/* Resources List Section */}
<Card className="shadow-lg border border-gray-100 dark:border-gray-700">
  <CardHeader className="flex flex-row items-center justify-between pb-4">
    <div>
      <CardTitle className="text-2xl font-bold flex items-center gap-2">
        <FileTextIcon className="h-6 w-6 text-blue-600 dark:text-blue-400" /> All Shared Resources
      </CardTitle>
      <CardDescription>Browse and download documents shared by colleges across the
network.</CardDescription>
    </div>
    {/* Filters & Search */}
    <div className="flex items-center space-x-2">
      <div className="relative">
        <SearchIcon className="absolute left-3 top-1/2 -translate-y-1/2 h-4 w-4 text-muted-foreground" />
        <Input
          type="text"
          placeholder="Search resources..."
          value={searchTerm}
          onChange={(e) => setSearchTerm(e.target.value)}
          className="pl-9 pr-3 w-48"
        />
      </div>
      <Select value={filterCategory} onValueChange={setFilterCategory}>
        <SelectTrigger className="w-[180px]">
          <FilterIcon className="mr-2 h-4 w-4" />
          <SelectValue placeholder="Filter by Category" />
        </SelectTrigger>
      </Select>
    </div>
  </CardHeader>
  <CardContent>
    <Toaster />
  </CardContent>
</Card>
);
}; export default ResourcesPage;

```

**Colleges Page:**

```
return (  
  <div className="container mx-auto py-8 px-4 md:px-6 lg:px-8">  
    <h1 className="text-4xl font-extrabold text-gray-800 dark:text-gray-100 mb-8 text-center">  
      Explore Other <span className="text-blue-600 dark:text-blue-400">Colleges</span>  
    </h1>  
  
    <Card className="mb-8 shadow-lg border border-gray-100 dark:border-gray-700">  
      <CardHeader className="flex flex-row items-center justify-between pb-4">  
        <CardTitle className="text-2xl font-bold flex items-center gap-2">  
          <UsersIcon className="h-6 w-6 text-blue-600 dark:text-blue-400" /> College Directory  
        </CardTitle>  
        <div className="relative w-full max-w-sm">  
          <SearchIcon className="absolute left-3 top-1/2 -translate-y-1/2 h-4 w-4 text-muted-foreground" />  
          <Input  
            type="text"  
            placeholder="Search colleges by name, city..."  
            value={searchTerm} // Now controlled by local state, which is updated from URL  
            onChange={(e) => setSearchTerm(e.target.value)} // Allows typing directly on this page  
            className="w-full pl-9 pr-4 rounded-md"  
          />
```

```

</div>
</CardHeader>
<CardContent>
  {isLoadingColleges ? (
    <div className="flex justify-center items-center h-48">
      <Loader2 className="h-8 w-8 animate-spin text-blue-600 dark:text-blue-400" />
      <p className="ml-2 text-lg text-gray-600 dark:text-gray-300">Loading colleges...</p>
    </div>
  ) : filteredColleges.length === 0 ? (
    <div className="text-center py-10 text-gray-500 dark:text-gray-400">
      <p className="text-lg">No other colleges found matching your criteria.</p>
      <p className="text-sm mt-2">Try adjusting your search or check back later!</p>
    </div>
  ) : (
    <div className="grid grid-cols-1 md:grid-cols-2 lg:grid-cols-3 gap-6">
      {filteredColleges.map(college => (
        <Card key={college._id} className="shadow-md hover:shadow-lg transition-shadow duration-200 flex flex-col justify-between border border-gray-200 dark:border-gray-700">
          <CardHeader className="pb-3 flex-row items-center space-x-3">
            <Avatar className="h-10 w-10 border-2">
              {college.logo ? (
                <AvatarImage src={constructLogoUrl(college.logo)} alt={` ${college.name} Logo`} />
              ) : (
                <AvatarFallback className="bg-primary text-primary-foreground text-md font-semibold">{college.name?.charAt(0) || 'C'}</AvatarFallback>
              )}
            </Avatar>
            <div className="flex-1">
              <CardTitle className="text-lg font-semibold truncate">{college.name}</CardTitle>
              <CardDescription className="text-xs text-muted-foreground">
                <MapPinIcon className="inline-block h-3 w-3 mr-1" />
                {[college.city, college.state, college.country].filter(Boolean).join(', ')}
              </CardDescription>
            </div>
          </CardHeader>
          <CardContent className="flex-1 text-sm text-gray-700 dark:text-gray-300 pb-3 space-y-2">
            <p className="line-clamp-3">{college.description}</p>
            {college.website && (
              <a href={college.website} target="_blank" rel="noopener noreferrer" className="flex items-center gap-1 text-xs text-blue-600 hover:underline dark:text-blue-400">
                <GlobeIcon className="h-3 w-3" /> Visit Website
              </a>
            )}
          </CardContent>
        </div>
      ) : (
        <div className="p-4 border-t flex justify-end items-center bg-muted/20">
          <Button

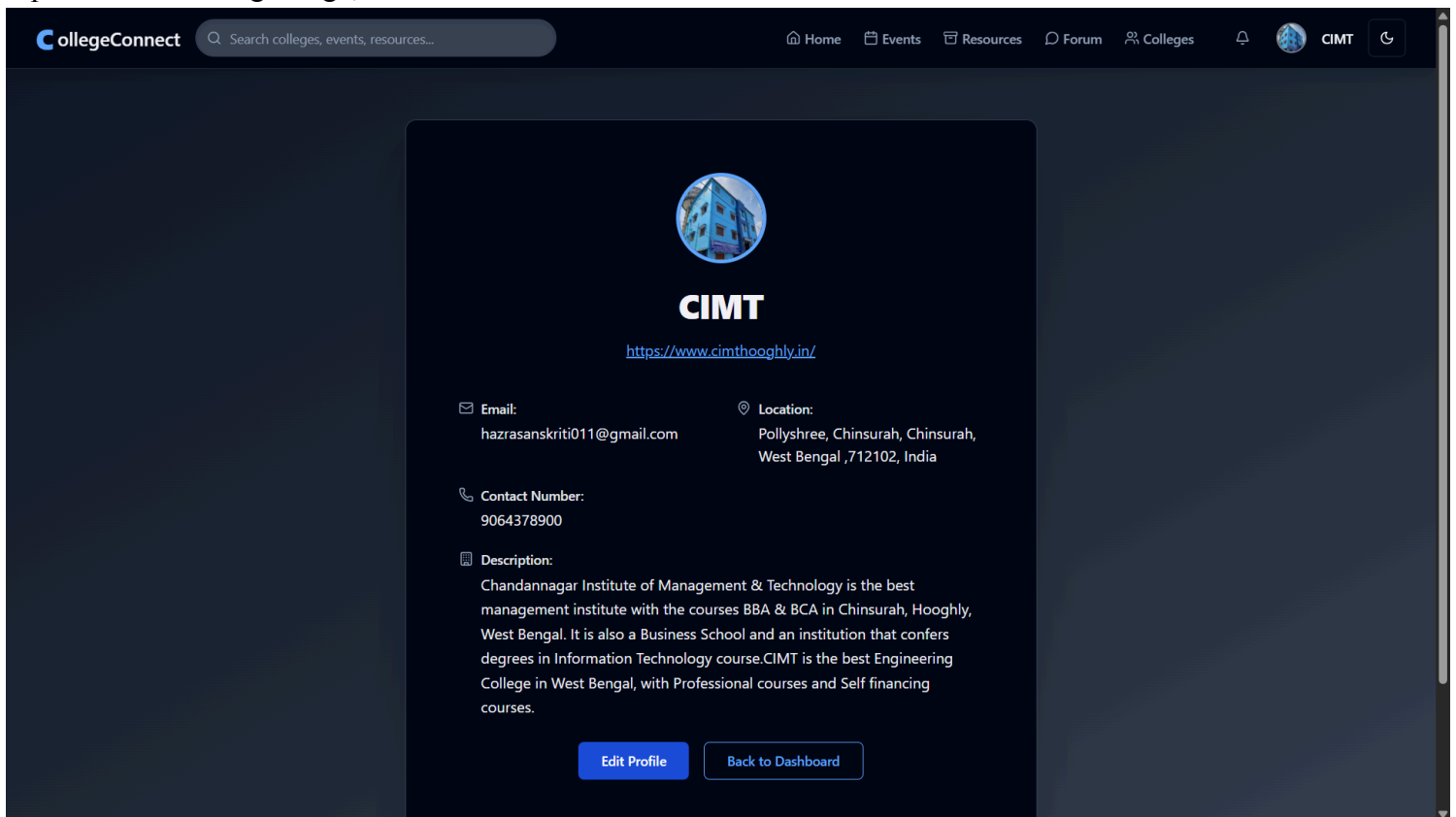
```

```

        variant="outline"
        size="sm"
        onClick={() => navigate(`/colleges/${college._id}`)}
        className="flex items-center gap-1 text-blue-600 hover:text-blue-700 dark:text-blue-400
dark:hover:text-blue-300"
      >
        <EyeIcon className="h-4 w-4" /> View Profile
      </Button>
    </div>
  </Card>
)}
</div>
)}
</CardContent>
</Card>
<Toaster />
</div>
);
};

```

export default CollegesPage;



College Profile:

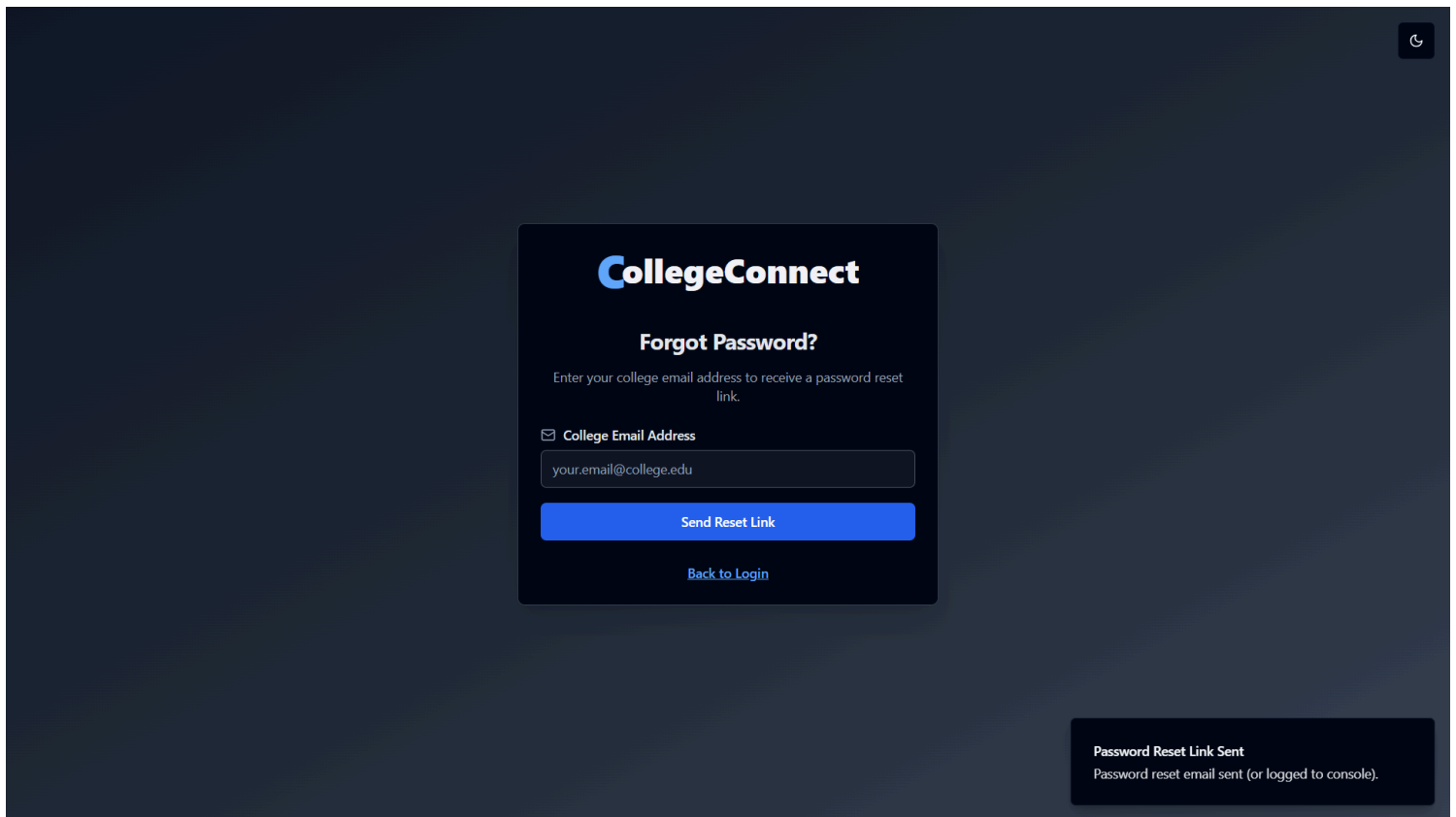
```
return (  
  <div className="min-h-screen flex items-center justify-center bg-gradient-to-br from-blue-200 via-purple-200  
to-blue-300 dark:from-gray-900 dark:via-gray-800 dark:to-gray-700 p-4 sm:p-6 lg:p-8">  
    <Card className="w-full max-w-2xl mx-auto shadow-2xl rounded-xl border border-gray-100  
dark:border-gray-700 p-6 md:p-8">  
      <CardHeader className="text-center space-y-4 pb-6">  
        { /* Circular Logo Display */ }  
        <Avatar className="w-24 h-24 mx-auto mb-2 border-4 border-blue-600 dark:border-blue-400 shadow-md">  
          {collegeLogoUrl ? (  
            <AvatarImage src={collegeLogoUrl} alt={` ${profileData.name} Logo` } />  
          ) : (  
            <AvatarFallback className="text-3xl font-semibold bg-blue-100 dark:bg-gray-800 text-blue-600  
dark:text-blue-400"> {profileData.name.charAt(0)} </AvatarFallback>  
          )}  
        </Avatar>  
  
        <CardTitle className="text-4xl font-extrabold text-gray-800  
dark:text-gray-100"> {profileData.name} </CardTitle>  
        { /* Website just below the name */ }  
        <CardDescription className="text-md text-gray-600 dark:text-gray-300">  
          {profileData.website ? (  
            <a href={profileData.website} target="_blank" rel="noopener noreferrer" className="underline  
text-blue-600 hover:text-blue-700 dark:text-blue-400 dark:hover:text-blue-300 transition-colors duration-200">  
              {profileData.website}  
            </a>  
          ) : 'N/A'}  
        </CardDescription>  
      </CardHeader>  
      <CardContent className="space-y-6 pt-4">  
        <div className="grid grid-cols-1 md:grid-cols-2 gap-x-8 gap-y-5 text-left">  
          { /* Email in the list */ }  
          <div className="space-y-1">  
            <Label className="text-sm font-semibold text-gray-700 dark:text-gray-200 flex items-center gap-2">  
              <MailIcon className="h-4 w-4 text-muted-foreground" /> Email:  
            </Label>  
            <p className="text-base text-gray-800 dark:text-gray-100 pl-6"> {profileData.email} </p>  
          </div>  
          { /* Location */ }  
          <div className="space-y-1">  
            <Label className="text-sm font-semibold text-gray-700 dark:text-gray-200 flex items-center gap-2">  
              <MapPinIcon className="h-4 w-4 text-muted-foreground" /> Location:  
            </Label>  
            <p className="text-base text-gray-800 dark:text-gray-100 pl-6">  
              { [profileData.address, profileData.city, profileData.state, profileData.country]  
                .filter(Boolean)  
              }  
            </p>  
          </div>  
        </div>  
      </CardContent>  
    </div>  
  )  
);
```

```

        .join(', ') || 'N/A'}
    </p>
</div>
{/* Contact Number */}
<div className="space-y-1">
    <Label className="text-sm font-semibold text-gray-700 dark:text-gray-200 flex items-center gap-2">
        <PhoneIcon className="h-4 w-4 text-muted-foreground" /> Contact Number:
    </Label>
    <p className="text-base text-gray-800 dark:text-gray-100 pl-6">{profileData.contactNumber ||
'N/A'}</p>
</div>
{/* Description takes full width on md and up */}
<div className="space-y-1 col-span-1 md:col-span-2">
    <Label className="text-sm font-semibold text-gray-700 dark:text-gray-200 flex items-center gap-2">
        <BuildingIcon className="h-4 w-4 text-muted-foreground" /> Description:
    </Label>
    <p className="text-base text-gray-800 dark:text-gray-100 leading-relaxed
pl-6">{profileData.description}</p>
</div>
</div>
{/* Edit Profile and Back to Dashboard Buttons */}
<div className="text-center mt-8 space-x-4">
    <Button
        onClick={() => navigate('/edit-profile')}
        className="px-6 py-2 bg-blue-600 hover:bg-blue-700 dark:bg-blue-700 dark:hover:bg-blue-600
text-white font-semibold rounded-md shadow-md transition-colors duration-200 ease-in-out"
    >
        Edit Profile
    </Button>
    <Button
        onClick={() => navigate('/') }
        variant="outline"
        className="px-6 py-2 border-blue-600 text-blue-600 hover:bg-blue-50 dark:border-blue-400
dark:text-blue-400 dark:hover:bg-gray-700 font-semibold rounded-md shadow-md transition-colors duration-200
ease-in-out"
    >
        Back to Dashboard
    </Button>
</div>
</CardContent>
</Card>
<Toaster />
</div>
);
};

```

export default CollegeProfilePage;



ForgotPassword Page:

```
return (  
  <div className="min-h-screen flex items-center justify-center bg-gradient-to-br from-blue-200 via-purple-200  
to-blue-300 dark:from-gray-900 dark:via-gray-800 dark:to-gray-700 p-4 relative">  
    <div className="absolute top-4 right-4 z-10">  
      <ModeToggle />  
    </div>  
    <Card className="w-full max-w-md mx-auto shadow-xl rounded-lg border border-gray-100  
dark:border-gray-700">  
      <CardHeader className="space-y-3 text-center">  
        <div className="flex items-center justify-center mb-6">  
          <span className="text-blue-600 dark:text-blue-400 text-5xl font-extrabold -mr-1">C</span>  
          <h1 className="text-4xl font-extrabold text-gray-800 dark:text-gray-100">ollegeConnect</h1>  
        </div>  
        <CardTitle className="text-2xl font-bold text-gray-700 dark:text-gray-200 mt-2">Forgot  
Password?</CardTitle>  
        <CardDescription className="text-gray-600 dark:text-gray-400">Enter your college email address to  
receive a password reset link.</CardDescription>  
      </CardHeader>  
      <CardContent>
```



```

<form onSubmit={handleSubmit} className="space-y-4">
  <div className="grid gap-2">
    <Label htmlFor="email" className="text-gray-700 dark:text-gray-200 flex items-center gap-2">
      <MailIcon className="h-4 w-4 text-muted-foreground" /> College Email Address
    </Label>
    <Input
      id="email"
      type="email"
      placeholder="your.email@college.edu"
      value={email}
      onChange={(e) => setEmail(e.target.value)}
      required
      className="bg-white dark:bg-gray-900 border-gray-300 dark:border-gray-600 focus:ring-blue-500
focus:border-blue-500 rounded-md py-2 px-3"
    />
  </div>
  <Button
    type="submit"
    className="w-full bg-blue-600 hover:bg-blue-700 dark:bg-blue-700 dark:hover:bg-blue-600 text-white
font-semibold py-2 rounded-md transition-colors duration-200 ease-in-out"
    disabled={isSubmitting}
  >
    {isSubmitting ? <Loader2 className="mr-2 h-4 w-4 animate-spin" /> : 'Send Reset Link'}
  </Button>
</form>
<div className="mt-6 text-center text-sm">
  <Link to="/login" className="text-blue-600 hover:text-blue-700 dark:text-blue-400
dark:hover:text-blue-300 font-medium transition-colors duration-200 ease-in-out underline">
    Back to Login
  </Link>
</div>
</CardContent>
</Card>
<Toaster />
</div>
);
};

export default ForgotPasswordPage;

```

Reset Password:

CollegeConnect

Reset Password

Enter your new password below.

New Password

Enter new password

Confirm New Password

Confirm new password

Reset Password

Notifications :

CollegeConnect

Search colleges, events, resources...

Home

Events

Resources

Forum

Colleges

3

Vivekananda Mahavidyalaya

Notifications

New Resource: "Python EBook" shared by CIMT.
from CIMT
6/20/2025, 6:08:10 PM

New Forum Post: " Academic & Career-Oriented Topics " by CIMT.
from CIMT
6/20/2025, 5:53:59 PM

New Event: "Inter-College Sports Event" by CIMT on 26/6/2025.
from CIMT
6/20/2025, 5:49:33 PM

Mark all as read

Welcome to CollegeConnect
Vivekananda Mahavidyalaya

This is your central dashboard. Explore recent activities, shared resources, and active discussions from colleges in your network.

Recent Activities & Updates

Stay informed with the latest events, newly shared resources, and active discussions from colleges across the platform. This section will soon feature a dynamic feed of real-time collaboration.

Browse Colleges

View All Events

Your gateway to inter-college collaboration and communication. Streamline your processes, share knowledge, and foster a connected academic community.

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5B. BACKEND

db.js:

```
const mongoose = require('mongoose');

const connectDB = async () => {
  try {
    const conn = await mongoose.connect(process.env.MONGO_URI);
    console.log('MongoDB Connected (Local): ${conn.connection.host}');
  } catch (error) {
    console.error(`Error connecting to MongoDB: ${error.message}`);
    process.exit(1); // Exit process with failure
  }
};

module.exports = connectDB;
```

server.js:

```
require('dotenv').config();
const express = require('express');
const connectDB = require('./config/db');
const cors = require('cors');
const authRoutes = require('./routes/authRoutes');
const collegeRoutes = require('./routes/collegeRoutes'); // NEW: Import college routes
const resourceRoutes = require('./routes/resourceRoutes'); // NEW: Import resource routes
const notificationRoutes = require('./routes/notificationRoutes');
const eventRoutes = require('./routes/eventRoutes');
const forumRoutes = require('./routes/forumRoutes');

const path = require('path');

const app = express();

// Connect to database
connectDB();

// Middleware
app.use(express.json());
app.use(cors());
app.use('/uploads', express.static(path.join(__dirname, 'uploads')));
app.use('/uploads/resources', express.static(path.join(__dirname, 'uploads', 'resources')));

// Define API routes
app.use('/api/auth', authRoutes);
app.use('/api/colleges', collegeRoutes); // NEW: Mount college routes
app.use('/api/resources', resourceRoutes); // NEW: Mount resource routes
app.use('/api/notifications', notificationRoutes); // NEW: Mount notification routes
app.use('/api/events', eventRoutes);
app.use('/api/forum', forumRoutes);
```

```
// Basic route to check if server is running
app.get('/', (req, res) => {
  res.send('API is running...');
});

// Error handling middleware
app.use((err, req, res, next) => {
  console.error(err.stack);
  const statusCode = res.statusCode === 200 ? 500 : res.statusCode;
  res.status(statusCode);
  res.json({
    message: err.message,
    stack: process.env.NODE_ENV === 'production' ? null : err.stack,
  });
});

const PORT = process.env.PORT || 5000;

app.listen(PORT, () => console.log(`Server running on port ${PORT}`));
```

Models:

college.js:

```
const mongoose = require('mongoose');
const bcrypt = require('bcryptjs');
const crypto = require('crypto');

const collegeSchema = mongoose.Schema(
  {
    name: {
      type: String,
      required: true,
      unique: true, // College names should ideally be unique
    },
    email: { // Contact email for the college
      type: String,
      required: true,
      unique: true,
    },
    password: { // Password for the college's login
      type: String,
      required: true,
    },
    address: {
      type: String,
      required: true,
    },
    city: {
      type: String,
      required: true,
    },
  }
);
```

```

    },
    state: {
      type: String,
      required: true,
    },
    country: {
      type: String,
      required: true,
    },
    description: {
      type: String,
      required: true,
      maxlength: 1000, // Limit description length
    },
    logo: { // Path to the uploaded logo image
      type: String,
      default: '/uploads/default_college_logo.png', // Default logo if none uploaded
    },
    website: {
      type: String,
      required: false, // Optional
    },
    contactNumber: {
      type: String,
      required: false, // Optional
    },
    resetPasswordToken: String, // NEW: Field to store the reset token
    resetPasswordExpire: Date,
    // You can add more fields here later, e.g., verified status, faculty, etc.
  },
  {
    timestamps: true, // Adds createdAt and updatedAt fields automatically
  }
);

// Middleware to hash password before saving
collegeSchema.pre('save', async function (next) {
  if (!this.isModified('password')) { // Only hash if password was modified
    next();
  }

  const salt = await bcrypt.genSalt(10);
  this.password = await bcrypt.hash(this.password, salt);
  next();
});

```

```

// Method to compare entered password with hashed password in database
collegeSchema.methods.matchPassword = async function (enteredPassword) {
  return await bcrypt.compare(enteredPassword, this.password);
}

```

```

};

// Method to generate and hash password reset token
collegeSchema.methods.getResetPasswordToken = function () {
  // Generate token (raw, non-hashed)
  const resetToken = crypto.randomBytes(20).toString('hex'); // Requires 'crypto' module

  // Hash token and set to resetPasswordToken field
  // This is important: Store the HASHED token in DB for security
  this.resetPasswordToken = crypto.createHash('sha256').update(resetToken).digest('hex');

  // Set token expire time (e.g., 10 minutes from now)
  this.resetPasswordExpire = Date.now() + 10 * 60 * 1000; // 10 minutes in milliseconds

  return resetToken; // Return the UNHASHED token to send in email
};

const College = mongoose.model('College', collegeSchema);

module.exports = College;

```

Controller:

authcontroller.js:

```

const asyncHandler = require('express-async-handler');
const College = require('../models/College'); // Changed from User to College
const generateToken = require('../utils/generateToken');
const nodemailer = require('nodemailer'); // NEW: Import nodemailer
const crypto = require('crypto');

// @desc   Register a new college
// @route  POST /api/auth/register
// @access Public
const registerCollege = asyncHandler(async (req, res) => {
  const { name, email, password, address, city, state, country, description, website, contactNumber } = req.body;
  const logo = req.file ? `uploads/${req.file.filename}` : undefined; // Get path to uploaded logo

  // Check if college already exists by email or name
  const collegeExists = await College.findOne({ $or: [{ email }, { name }] });

  if (collegeExists) {
    res.status(400); // Bad request
    throw new Error('College already exists with that email or name');
  }

  // Create new college
  const college = await College.create({
    name,
    email,
    password, // Password will be hashed by the pre-save middleware

```

```

    address,
    city,
    state,
    country,
    description,
    logo, // Include logo path if uploaded
    website,
    contactNumber,
  });

  if (college) {
    res.status(201).json({ // 201 Created
      _id: college._id,
      name: college.name,
      email: college.email,
      logo: college.logo, // Send back the logo path
      token: generateToken(college._id), // Generate JWT token for immediate login
    });
  } else {
    res.status(400);
    throw new Error('Invalid college data');
  }
});

// @desc   Authenticate college & get token
// @route   POST /api/auth/login
// @access  Public
const loginCollege = asyncHandler(async (req, res) => {
  const { email, password } = req.body;

  // Check for college by email
  const college = await College.findOne({ email });

  // Check if college exists and password matches
  if (college && (await college.matchPassword(password))) {
    res.json({
      _id: college._id,
      name: college.name,
      email: college.email,
      logo: college.logo, // Send back the logo path
      token: generateToken(college._id), // Generate JWT token for login
    });
  } else {
    res.status(401); // Unauthorized
    throw new Error('Invalid email or password');
  }
});

const forgotPassword = asyncHandler(async (req, res) => {
  const { email } = req.body;

```

```

const college = await College.findOne({ email });

if (!college) {
  res.status(404);
  throw new Error('College not found with that email address.');
```

}

```

// Generate reset token and set expiry
const resetToken = college.getResetPasswordToken(); // Method from College model

await college.save({ validateBeforeSave: false }); // Save college with new token, bypass validation for password field

// Create reset URL (This is the link sent to the user)
// Replace 'http://localhost:5173' with your actual frontend URL in production
const resetURL = `<span
class="math-inline">\{req.protocol\}\:\/</span>\{req.get('host')\}/reset-password/\${resetToken}`;
// IMPORTANT: The frontend URL is typically different from backend host.
// For development, use `http://localhost:5173/reset-password/\${resetToken}`
// For local testing, let's use a dummy URL for now.
const frontendResetURL = `http://localhost:5173/reset-password/\${resetToken}`;

// --- Email Sending Logic (Placeholder for now) ---
const message = `You are receiving this email because you (or someone else) has requested the reset of a password.
Please make a PUT request to: \n\n \${frontendResetURL} \n\n
This token is valid for 10 minutes. If you did not request this, please ignore this email.`;

try {
  // In a real application, you'd configure and use Nodemailer here
  // Example (requires email config in .env):

  const transporter = nodemailer.createTransport({
    service: 'Gmail', // or your email service
    auth: {
      user: process.env.EMAIL_USERNAME,
      pass: process.env.EMAIL_PASSWORD,
    },
  });

  // Inside forgotPassword function:
  const mailOptions = {
    from: process.env.EMAIL_USERNAME, // Use your Gmail address from .env
    to: college.email,
    subject: 'Password Reset Request',
    text: message,
  };

```



```

await transporter.sendMail(mailOptions);

console.log('--- PASSWORD RESET LINK (FOR TESTING) ---');
console.log(`To: ${college.email}`);
console.log(`Subject: Password Reset Request for ${college.name}`);
console.log(`Link: ${frontendResetURL}`);
console.log('--- END RESET LINK ---');

res.status(200).json({ success: true, data: 'Password reset email sent (or logged to console).' });
} catch (err) {
  // If email fails, clear token and expiry so user can request again
  college.resetPasswordToken = undefined;
  college.resetPasswordExpire = undefined;
  await college.save({ validateBeforeSave: false });

  console.error('Error sending reset email:', err);
  res.status(500);
  throw new Error('Error sending password reset email. Please try again later.');
```

```

}
});

```

```

// @desc   Reset password (using token)
// @route  PUT /api/auth/reset-password/:token
// @access Public
const resetPassword = asyncHandler(async (req, res) => {
  // Hash the token from the URL parameter to match the one stored in DB
  const resetPasswordToken = crypto.createHash('sha256').update(req.params.token).digest('hex');

  const college = await College.findOne({
    resetPasswordToken,
    resetPasswordExpire: { $gt: Date.now() }, // Check if token is not expired
  });

  if (!college) {
    res.status(400); // Bad request / Invalid token
    throw new Error('Invalid or expired password reset token.');
```

```

}

```

```

// Set new password
college.password = req.body.password; // The pre-save hook will hash it
college.resetPasswordToken = undefined; // Clear token after use
college.resetPasswordExpire = undefined; // Clear expiry

```

```

await college.save(); // Save new password and clear token

```

```

res.status(200).json({ success: true, message: 'Password reset successfully. You can now log in.' });
});

```

```
module.exports = { registerCollege, loginCollege, forgotPassword,resetPassword }; // Export new function names
```

Routes:

collegeroutes.js

```
const express = require('express');
const { getMyCollegeProfile, updateMyCollegeProfile, getCollegeById, getAllColleges } =
require('../controllers/collegeController'); // NEW: Import update function
const { protect } = require('../middleware/authMiddleware');
const multer = require('multer'); // NEW: Import multer
const path = require('path'); // NEW: Import path

const router = express.Router();

// Configure Multer for file storage (re-use from authRoutes, or define here again for clarity)
// It's generally better to define it here if it's specific to these routes
const storage = multer.diskStorage({
  destination: function (req, file, cb) {
    cb(null, 'uploads/'); // Ensure this 'uploads' folder exists in backend root
  },
  filename: function (req, file, cb) {
    cb(null, `${file.fieldname}-${Date.now()}${path.extname(file.originalname)}`);
  }
});

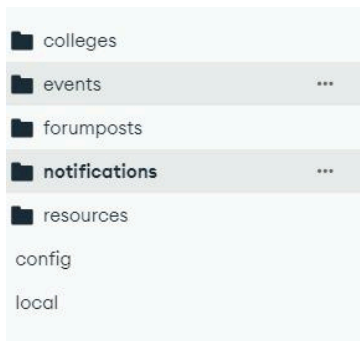
const fileFilter = (req, file, cb) => {
  if (file.mimetype.startsWith('image/')) {
    cb(null, true);
  } else {
    cb(new Error('Only image files are allowed!'), false);
  }
};

const upload = multer({
  storage: storage,
  fileFilter: fileFilter,
  limits: { fileSize: 5 * 1024 * 1024 } // 5MB limit
});

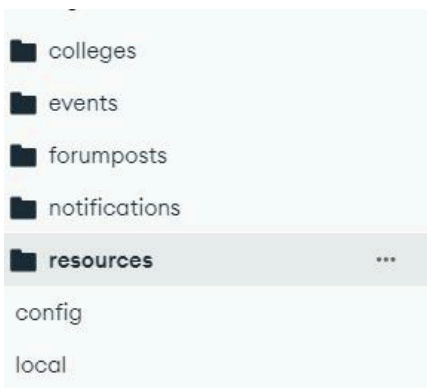
// Protected routes
router.route('/me')
  .get(protect, getMyCollegeProfile) // Get logged-in college's profile
  .put(protect, upload.single('logo'), updateMyCollegeProfile); // NEW: Update logged-in college's profile

router.route('/:id').get(protect, getCollegeById); // Get specific college's profile by ID
router.route('/').get(protect, getAllColleges); // Get all colleges (for directory/feed)
module.exports = router;
```

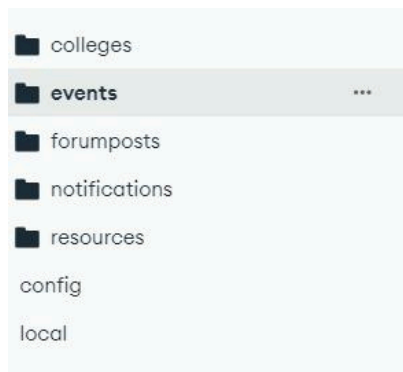
5C. DATABASE SCHEMA



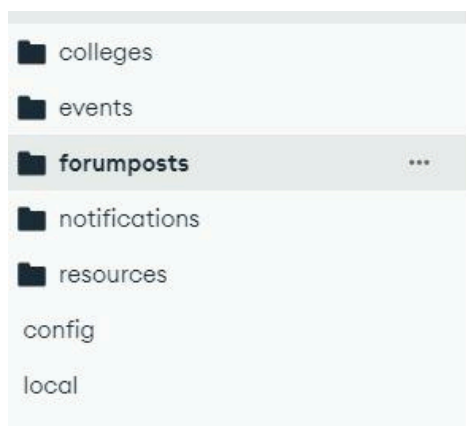
```
_id: ObjectId('6834204f090f4763ff103c90')
recipient: ObjectId('68341d6e090f4763ff103bf9')
sender: ObjectId('68341ff1090f4763ff103c6a')
type: "new-resource"
message: "New Resource: "Football Tournament" shared by Vivekananda Mahavidyalay..."
read: false
link: "/resources/6834204f090f4763ff103c8d"
relatedId: ObjectId('6834204f090f4763ff103c8d')
__v: 0
createdAt: 2025-05-26T08:03:27.977+00:00
updatedAt: 2025-05-26T08:03:27.977+00:00
```



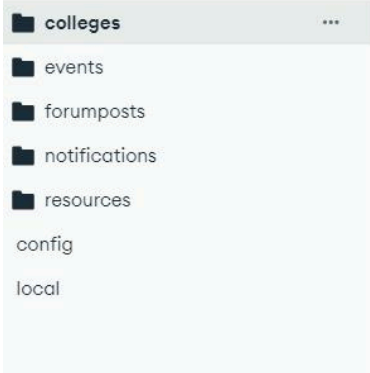
```
_id: ObjectId('68341e19090f4763ff103c3a')
title: "New Syllabus update"
description: "Notice regarding new Syllabus update"
category: "Reports & Academic Content"
file: "/uploads/resourceFile-1748246041465.png"
originalFileName: "Screenshot 2025-01-18 204543.png"
uploadedBy: ObjectId('68341d6e090f4763ff103bf9')
createdAt: 2025-05-26T07:54:01.467+00:00
updatedAt: 2025-05-26T07:54:01.467+00:00
__v: 0
```



```
_id: ObjectId('683f0d0f5dfd698709f4fd17')
title: "Techtrix"
description: "Basically a tech Competition"
eventType: "Workshop"
location: "67/A SC Road, Kolkata 700001"
dateTime: 2025-06-11T05:00:00.000+00:00
organizingCollege: ObjectId('68341d6e090f4763ff103bf9')
contactEmail: "CIMT@gmail.com"
contactPhone: "9856231470"
registrationLink: "https://TechTrix.com"
createdAt: 2025-06-03T14:56:15.843+00:00
updatedAt: 2025-06-03T14:56:15.843+00:00
__v: 0
```



```
_id: ObjectId('683f103c5ffe248be179391a')
title: "test title"
content: "test test"
postedBy: ObjectId('68341d6e090f4763ff103bf9')
mentionedColleges: Array (empty)
createdAt: 2025-06-03T15:09:48.457+00:00
updatedAt: 2025-06-03T15:09:48.457+00:00
__v: 0
```



```
_id: ObjectId('68341d6e090f4763ff103bf9')
name: "Chandannagar Institute Of Management & Technology"
email: "cimt@gmail.com"
password: "$2b$10$sDZW5iIvwDUtYKCgP1Zo20R1AG6LPyyJwycQ4w.IQQd6oE032ncTG"
address: "V9RH+627, Pollyshree, Chinsurah Station Road"
city: "Chinsurah, Chinsurah"
state: "West Bengal 712102"
country: "India"
description: "CIMT College, Canadian Institute of Management and Technology - Regist..."
logo: "/uploads/logo-1748245870007.jpeg"
website: "https://www.cimtcollege.com/cimt/"
contactNumber: "9876543289"
createdAt: 2025-05-26T07:51:10.041+00:00
updatedAt: 2025-05-26T08:04:58.316+00:00
```

6. CONCLUSION

CollegeConnect serves as a powerful tool for enhancing inter-institutional communication and collaboration, making academic and administrative interactions more accessible, streamlined, and interconnected. By leveraging technology, it bridges the gap between different educational institutions, enabling flexible and efficient information exchange and partnership building anytime, anywhere. The platform not only supports a wide range of communication needs—from resource sharing and event coordination to forum discussions and directory Browse—but also incorporates features like real-time notifications and profile management to enrich institutional engagement. As digital collaboration continues to grow in the academic sector, CollegeConnect stands as a promising solution to meet the evolving communication and cooperative needs of colleges and universities alike.

7. FUTURE SCOPE & FURTHER ENHANCEMENTS

❖ Future scope:-

The future of CollegeConnect is promising, with numerous opportunities for growth and innovation in fostering inter-institutional communication and collaboration. Key areas for development include:

1. **AI-Powered Personalization:** Integrating artificial intelligence to offer personalized feeds of relevant resources and events, smart recommendations for potential collaborative partner colleges, and adaptive suggestions for inter-college initiatives based on institutional profiles and past interactions.
2. **Gamification:** Adding more interactive and game-like features to boost college engagement, such as leaderboards for most active contributors (e.g., resources shared, events hosted), badges for successful collaborations, or challenge systems for inter-college competitions.
3. **AR/VR Integration:** Enhancing inter-college engagement with augmented and virtual reality for immersive experiences, such as virtual campus tours for student exchange programs, virtual joint lab sessions, or immersive collaborative meeting spaces for inter-college project teams.
4. **Offline Accessibility:** Expanding features to allow content access without an internet connection, ensuring continued access to downloaded resources, cached event details, and forum discussions for review in remote or low-connectivity areas.
5. **Multilingual Support:** Supporting multiple languages to cater to a diverse, global academic user base, facilitating partnerships with international institutions.
6. **Advanced Collaborative Tools:** Incorporating more sophisticated real-time collaboration features like dedicated project-specific chat rooms (addressing the earlier chat discussion), shared document editing, advanced group project management tools, and peer review mechanisms for joint academic endeavors.

❖ Further enhancement:-

The future of CollegeConnect holds immense potential for advanced features and broader integration, continuously enhancing inter-institutional communication and collaboration. Key areas for development include:

1. User Experience Enhancements:

- **Personalized Dashboards:** Implement AI-powered dashboards to show recent activities from connected colleges, tailored suggestions for resources or events, and quick overviews of a college's own contributions and ongoing collaborations.
- **Gamification for Collaboration:** Introduce badges for active participation (e.g., resources shared, events hosted, successful partnerships), leaderboards for highly collaborative colleges, and achievement systems to motivate deeper engagement.
- **Enhanced Accessibility Features:** Beyond the existing Dark Mode, improve usability for all users by integrating advanced accessibility features such as comprehensive screen reader support for complex interfaces and robust keyboard navigation.

2. Collaboration & Information Tools:

- **Interactive Forms & Task Management:** Develop interactive forms for structured collaborative inquiries or project proposals, and incorporate task management features with checklists for joint inter-college initiatives.
- **Event Content Enrichment:** Allow users to take private notes on shared event recordings or webinars hosted on the platform, and include auto-generated transcripts for easy review.
- **AI-Powered Assistance:** Integrate AI chatbots to provide on-demand assistance for drafting inter-college proposals, suggesting best practices for collaborative projects, or answering FAQs about platform usage.

3. Community & Interaction Features:

- **Integrated Live Sessions:** Incorporate real-time video conferencing tools to host inter-college webinars, facilitate joint virtual meetings for collaborative projects, or enable live Q&A sessions with expert faculty from different institutions.
- **Structured Collaboration Matching:** Develop features to match colleges for specific collaborations (e.g., research partnerships, student exchange programs), or create specialized interest groups for academic disciplines.

4. Analytics & Progress Tracking (Institutional Level):

- **Detailed Engagement Analytics:** Provide dashboards for college administrators offering insights into their institution's platform engagement, including resource download rates, event participation from other colleges, and forum activity trends.
- **Adaptive Collaboration Pathways:** Implement a recommendation system that suggests collaborative opportunities or content based on a college's demonstrated interests, expertise, and historical interaction patterns.

5. Technical Integrations:

- **Offline Access & Cloud Sync:** Allow offline access to previously downloaded resources, cached event schedules, and forum discussions, ensuring continuity of information even without an internet connection.
- **Multilingual Support:** Expand the platform's reach globally by providing multilingual UI and content support to facilitate international academic partnerships.
- **ERP/LMS Integration:** Explore integration with existing College ERP systems, Learning

Management Systems (LMS) like Moodle or Canvas, or other institutional portals for seamless data exchange and notification synchronization

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